



Sludge amended soil induced multidrug and heavy metal resistance in endophytic *Exiguobacterium* sp. E21L: genomics evidences

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Abstract

The emergence of multidrug-resistant bacteria in agro-environments poses serious risks to public health and ecological balance. In this study, *Exiguobacterium* sp. E21L, an endophytic strain, was isolated from carrot leaves cultivated in soil amended with sewage treatment plant-derived sludge. The strain exhibited resistance to clinically relevant antibiotics, including beta-lactams, fluoroquinolones, aminoglycosides, and macrolides, with a high Multi-Antibiotic Resistance Index of 0.88. Whole-genome sequencing revealed a genome of 3.06 Mb, encoding 3894 protein-coding genes, including antimicrobial resistance genes (ARGs) such as *blaNDM*, *ermF*, *tetW*, and *sulI*, along with heavy metal resistance genes (HMRGs) like *czcD*, *copB*, and *nikA*. Genomic islands carrying ARGs and stress-related genes suggested potential horizontal gene transfer. The strain demonstrated robust biofilm formation, high cell hydrophobicity (> 80%), and significant auto-aggregation (90% at 48 h), correlating with genes associated with motility, quorum sensing, and stress adaptation. Notably, phenotypic assays confirmed survival under simulated gastrointestinal conditions, emphasizing its resilience in host-associated environments. Comparative genomics positioned *Exiguobacterium* sp. E21L near *Exiguobacterium chiriquicha* RW-2, with a core genome of 2716 conserved genes. Functional annotations revealed genes involved in xenobiotic degradation, multidrug efflux pumps, and ABC-type transporters, indicating versatile resistance mechanisms and metabolic capabilities. The presence of ARGs, HMRGs, and MGEs (mobile genetic elements) highlights the potential role of *Exiguobacterium* sp. E21L as a reservoir for resistance determinants in agricultural ecosystems. These findings emphasized the need for stringent regulations on sludge-based fertilizers and advanced sludge treatment strategies to mitigate AMR risks in agro-environments.

Keywords *Exiguobacterium* sp. E21L · Multidrug-resistant endophyte · Sewage sludge · Whole genome sequencing · Co-selection

Introduction

The rise of AMR represents a critical challenge to both public health and economic stability worldwide, with estimates suggesting that associated healthcare costs may surpass \$1.4 trillion USD by 2050. In 2019, bacterial infections resistant to treatment were directly responsible for approximately 1.27 million fatalities, exceeding the total deaths caused by

HIV/AIDS and malaria combined. (Murray et al. 2022; Thompson et al. 2023). Each year, an estimated 7.7 million deaths annually are attributed to bacterial infections, and 4.95 million are linked to drug-resistant pathogens, underscoring the urgent need for interventions (Ikuta et al. 2022). AMR develops as a natural adaptive response of bacteria to their surroundings; nevertheless, the widespread and intensive use of antibiotics over the last century has significantly hastened the emergence and spread of ARGs due to selective pressures in various ecological niches (Rolff et al. 2024). A study by Klein et al. (2018) across 76 countries revealed a 90.9% rise in the consumption of WHO-designated “Watch” antibiotics between 2000 and 2015, especially in low-middle-income nations. Concurrently, resistance among disease-causing bacteria has surged, complicating the treatment of infections with existing antibiotics (Aslam et al. 2018). Recently, AMR has been declared a global crisis by

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the World Health Organization (WHO), necessitating immediate action to combat this “silent pandemic” (Paneri and Sevta 2023).

Environmental reservoirs, such as sewage treatment plants (STPs), significantly contribute to the emergence and dissemination of ARGs and MGEs (Zhang et al. 2024). Unmetabolized antibiotics, heavy metals, microplastics, and biocides in sewage create stressors that promote bacterial resistance through mutation and HGT (Murray 2024). These processes enable bacteria to withstand antimicrobial selection pressures, including antibiotics, metals, and other pollutants, facilitating the emergence of multidrug resistance (Christaki et al. 2020). Resistance genes often reside in commensal bacteria, which can transfer these genes to pathogenic hosts, posing risks to humans and animals through environmental and food chain exposure (Djordjevic et al. 2024). This environmental dissemination of AMR amplifies the risk of human exposure and complicates efforts to mitigate the spread of resistance, as it involves intricate ecological and molecular interactions that demand further investigation (Wernli et al. 2022). While STP sludge is nutrient-rich and commonly used as fertilizer, it also serves as a reservoir for ARGs, contributing to their dissemination in agricultural settings (Patra et al. 2024a). Our previous studies demonstrated that sewage sludge amendments in agriculture can transfer antibiotic residues, toxic heavy metals and trace elements, which also influence the endophytic bacterial populations in crops such as spinach and carrot, leading to the identification of bacteria with high metal tolerance and elevated multi-antibiotic resistance index (MARI) scores (Patra and Dubey 2024; Patra et al. 2024b). These findings highlight the potential for AMR transmission via the food chain, especially when vegetables are consumed raw, as bacterial attachment to intestinal cells can enhance colonization and evade immune defences (Thomas et al. 2024). The gastrointestinal (GI) tract may function as a niche for resistant endophytic bacteria where bacterial attachment and evasion of immune defence enhance colonization (Ullah et al. 2025). Combined with virulence factors, AMR significantly increases bacterial survival in the GI environment, amplifying health risks. Additionally, the interaction of AMR with virulence traits further complicates the pathogenic potential of these bacteria as they adapt to survive harsh conditions in the host environment (Singh et al. 2024a, b). To investigate these dynamics further, we selected the endophytic bacterium *Exiguobacterium* sp. E21L for whole genome sequencing (WGS) analysis, aiming to elucidate its resistance traits.

Collins et al. (1983) initially introduced the genus *Exiguobacterium* to classify a group of coryneform bacteria characterized as sporogenous, non-acid-fast, irregularly structured Gram-positive bacilli. So far, 24 species have been identified under this genus (Keynan et al. 2007). A sister clade of the *Bacillus* genus, *Exiguobacterium* remains underexplored despite its notable adaptability to harsh environmental conditions, such

as elevated salinity, low temperatures, high radiation, and dehydration (Lopez et al. 2021; Xiao et al. 2024). This genus is frequently isolated from diverse habitats such as oceans, salt lakes, sediments, industrial wastewater, soil, glaciers, permafrost, atmospheric aerosols, and stromatolites and stromatolites highlighting its versatile ecological roles (Su et al. 2021; Zhang et al. 2021; Huo et al. 2023). Moreover, *Exiguobacterium* spp. have been identified in sludge-amended soil plant systems, with studies reporting their presence in the rhizosphere of plants such as *Brassica napus* in controlled pot experiments (Zhou et al. 2024) and in municipal waste-rich soil, where antibiotic- and heavy metal-resistant strains were isolated from the rhizospheres of *Coccinia cordifolia* and *Colocasia esculenta* (Gupta et al. 2012). These findings highlight the ecological adaptability of *Exiguobacterium* in sludge-impacted environments, reinforcing its potential role in AMR dissemination and heavy metal tolerance. Many *Exiguobacterium* species demonstrate resistance to heavy metals like chromium and arsenic through evolved detoxification mechanisms encoded in their genomes (da Costa et al. 2018; Kneubehl and Iyer 2023). Additionally, these bacteria exhibit potential for bioremediation, including degradation of synthetic polymers (Maroof et al. 2022; Sun et al. 2023), synthetic dyes (Kalpana et al. 2020; Haque et al. 2022), pesticides (Sur and Sathivelu 2024), and nitroaromatic chemicals (Arora et al. 2012). The genus has attracted growing attention for its environmental resilience and metabolic diversity, though its role in AMR dissemination remains underexplored. Although several studies have explored its evolutionary variation (Vishnivetskaya et al. 2009), energy processing mechanisms (Xiao et al. 2024), and industrial and agricultural applications (Marfetán et al. 2023; Kumari et al. 2024), there is a lack of targeted research on its AMR transmission potential. Furthermore, limited information exists on its pathogenicity.

Therefore, a detailed investigation of *Exiguobacterium* sp. is essential in understanding its role in environmental AMR dissemination and potential implications for clinical and ecological systems. This study seeks to address these gaps by examining *Exiguobacterium* sp. E21L, with the objectives to (a) characterize its genomic attributes and resistance determinants, (b) investigate its potential role as a multidrug-resistant endophyte influencing environmental ecosystems, (c) assess its contribution to risk assessment models, and (d) explore its adaptive strategies, with a focus on the risk of food chain contamination through plant-associated transmission.

Materials and methods

Isolation, detection and identification of the bacterial isolate

The bacterial strain *Exiguobacterium* sp. E21L was isolated from carrot leaf samples grown in soil amended with

sludge from the Dinapur STP (80 MLD) in Varanasi, India. The leaf samples were washed under tap water, air-dried, and then soaked in 20% hydrogen peroxide for 30 min to ensure sterility. Afterwards, they were rinsed thrice with sterile double distilled water for 3 min each. Subsequently, the samples were treated with 70% ethanol for 1 min and rewashed using the same procedure. Sterilization efficacy was verified by plating the final rinse water on LB agar and observing for any microbial growth. Sterilized leaf samples (3 g) were finely chopped, ground with sterilized mortar and pestle, and homogenized with 10 mL of sterile LB broth. Serial dilutions were performed up to 10^{-5} , and 100 μ L aliquots of each dilution were applied onto respective agar plates. These plates were kept at 28 ± 2 °C for 3 days to facilitate colony formation. Bacterial colonies exhibiting distinct morphological traits were isolated, purified on LB agar, and subjected to antibiogram profiling. Among these, strain E21L was selected for in-depth characterization based on its unique colony characteristics, high antibiotic resistance and heavy metal tolerance (0.5 to 1.5 mg mL⁻¹), and the relatively unexplored nature of *Exiguobacterium* spp. in the context of multidrug resistance. Glycerol stocks (50%) of the strain were prepared and stored at -80 °C to ensure long-term preservation. Genomic DNA from strain E21L strain was harvested from an overnight grown culture using the MasterPure Complete DNA and RNA Purification Kit according to the manufacturer's guidelines (Patra et al. 2024a). Finally, the strain was identified through 16S rRNA gene sequencing.

Characterization of the isolated strain

To investigate the growth dynamics of the E21L isolate under heavy metal stress, a loopful culture transferred into 100 mL of LB broth supplemented with different concentrations (1–5 mM) of heavy metal salts, including $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$, CdCl_2 , $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, $\text{K}_2\text{Cr}_2\text{O}_7$, $\text{MnSO}_4 \cdot \text{H}_2\text{O}$, $\text{Pb}(\text{NO}_3)_2$, and $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$. The conical flasks containing the cultures were kept in a shaker incubator set to 180 rpm for 24 h. Growth was monitored at 4 h intervals by measuring the absorbance at 600 nm (OD_{600}) using a spectrophotometer (The Science House, India). Antibiogram for E21L was determined through the Kirby-Bauer disk diffusion method, employing 17 antibiotics from diverse classes commonly used against environmental and clinical strains. These included aminoglycosides, beta-lactams, cephalosporins, chloramphenicol, fluoroquinolones, macrolides, polypeptides, rifamycin, sulfonamides, and tetracyclines. The evaluation followed standard protocols as described in Patra et al. (2024a), with resistance patterns analyzed using recommendations from the EUCAST (European Committee on

Antimicrobial Susceptibility Testing) and the CLSI (Clinical and Laboratory Standards Institute). Furthermore, the metabolic capabilities of strain E21L were assessed using the BIOLOG assay. A 100 μ L inoculum, prepared in Inoculation Fluid (IF), was added to a GEN III MicroPlate™ (Biolog Inc., Hayward, CA, USA) and the plate was kept at 37 °C for 24 h, following the protocol detailed by Mallick et al. (2018).

Amplification of ARGs, HMRGs, and MGEs using PCR

The selection of specific ARGs and HMRGs was based on the resistance and survival profiles of E21L against antibiotics and heavy metals. Genes were further selected considering the high accumulation of these contaminants in sludge-treated vegetables, suggesting potential resistance development in endophytes (Patra and Dubey 2024). A literature review also supported their relevance in plant-associated bacteria (Guo et al. 2021; Mandal et al. 2021; Chen et al. 2023), and conventional PCR was performed to confirm their presence before WGS. The analysis targeted a total of 13 ARGs, 2 MGEs, and 6 HMRGs, encompassing resistance to specific antibiotics. These included *blaCTX-M*, *blaGES*, and *blaTEM* (ampicillin); *blaNDM* (imipenem); *ermF* (erythromycin); *qnrS* (ciprofloxacin); *rpoB516*, *rpoB526*, and *rpoB531* (rifampicin); *Sul1* (cotrimoxazole and erythromycin); and *tetO*, *tetM*, and *tetW* (tetracycline). The MGEs analyzed were class 1 integron (*intI1*) and *IS26*. HMRGs targeted included *arsA* (arsenic), *copB* (copper), *cZcD* (cobalt, zinc, cadmium), *merA* (mercury), *nika* (nickel), and *silE* (copper/silver). Detailed information on gene-specific primers, including their sequences and expected amplicon sizes, is presented in Table S1. PCR amplification was conducted according to the methodology described in Patra and Dubey (2024), and the resulting products were separated on a 1.2% agarose gel to confirm the expected band sizes.

Assessment of in vitro pathogenic potential of *Exiguobacterium* sp. E21L

Adhesion and survival capability in the human gastrointestinal environment

This section highlights the *Exiguobacterium* sp. E21L strain, expanding upon our previous work that underscores the contribution of endophytes in the spread of AMR through the food chain (Patra and Dubey 2024). To gain deeper insights, a series of critical in vitro experiments were designed and performed to assess the colonization ability and survival potential of the E21L strain within the human gastrointestinal (GI) tract.

Cell surface hydrophobicity

The hydrophobic properties of *Exiguobacterium* sp. E21L were assessed following the methodology described by Kuebutornye et al. (2020). A bacterial suspension with an OD₆₀₀ adjusted to 1.0 was prepared and combined with an equal volume of various organic solvents, including hexane, hexadecane, ethyl acetate, and xylene. The mixture was thoroughly vortexed for 5 min, followed by incubation at 37 °C for 30 min to facilitate phase separation. The hydrophobicity (%) was calculated using the formula: $(1 - A_1/A_0) \times 100$, where A_0 is the OD₆₀₀ of the bacterial suspension before mixing, and A_1 is the OD₆₀₀ of the aqueous phase after phase separation.

Auto-aggregation assay

The auto-aggregation capability of the isolate was evaluated using the method outlined by Kuebutornye et al. (2020). Bacterial suspensions were prepared to an OD₆₀₀ of approximately 1.0 and transferred into test tubes, followed by vortexing for 30 s. The tubes were then incubated at room temperature. A_t designated time intervals (0, 2, 5, and 24 h), the supernatant was carefully collected, and its OD₆₀₀ was measured. The auto-aggregation percentage was calculated using the formula: $(1 - A_t/A_0) \times 100$, where A_0 represents the initial absorbance (0 h), and A_t denotes the absorbance at time t . Additionally, a visual examination of auto-aggregation was performed based on the procedure described by Patra et al. (2024a).

Static mono-compartmental model assay

Simulated GI conditions were created based on the methodology described by Abadia-García et al. (2013). Bacterial cells, adjusted to a turbidity corresponding to 0.04 at OD₆₀₀, were initially introduced into an artificial saliva medium containing 1.2 g L⁻¹ NaHCO₃, 0.22 g L⁻¹ CaCl₂, 2.2 g L⁻¹ KCl, and 6.2 g L⁻¹ NaCl. The suspension was maintained under batch conditions for 1 min at 37 °C and pH 7. Next, the bacteria were combined with a synthetic gastric solution prepared by dissolving 3 g L⁻¹ pepsin (sourced from porcine origin) in tryptic soy broth (TSB) with 0.5% NaCl and adjusting the pH to 2.0 using HCl. This mixture was incubated with continuous agitation at 37 °C for 3 h. The pH of the gastric medium was subsequently modified to 4.0 using 5% (v/v) hydrochloric acid. Following incubation, bacterial cells were collected by centrifugation at 10,000×g for 3 min and reconstituted in a synthetic small intestinal medium. This medium consisted of 1 g L⁻¹ pancreatin and 1.5 mg mL⁻¹ bile salts dissolved in sterile 0.5% NaCl solution, and the pH was adjusted to 8.0 using 0.1 M NaOH. The reconstituted bacteria were incubated at 37 °C for 4 h

under constant agitation. 100 µL of medium were collected at 30 min intervals and subsequently plated on LB agar and incubated at 37 °C for 48 h to determine colony-forming units (CFU). All the experiments were conducted in three replicates.

Evaluation of motility

The motility of the E21L isolate was evaluated using the swimming agar plate method. Stationary-phase cells of E21L (OD₆₀₀ ~ 1) were inoculated onto a semisolid motility agar composed of 0.3% agar, 0.5% sodium chloride, 1% tryptone, and 50 mg mL⁻¹ triphenyl tetrazolium chloride (TTC) dye. After incubation for 24 and 48 h, the motility zones were measured as described by Atkinson et al. (2006). Each assay was performed in triplicate for accuracy. A non-motile strain, *Staphylococcus aureus* MTCC 1430, was used as a negative control to validate the findings.

Qualitative and quantitative evaluation of biofilm formation

FESEM (Field emission scanning electron microscopy) was used to analyse the morphology of the biofilm and extracellular matrix produced by E21L cells, using a protocol adapted slightly from Pavez et al. (2023). Rectangular glass slides were placed in sterilized, flat-bottomed 6-well microtiter plates (MTP), each containing 4 mL of Luria–Bertani (LB) broth, with or without vancomycin supplementation (30 µg mL⁻¹). E21L cells, adjusted to an OD₆₀₀ of approximately 0.5, were inoculated into the wells and incubated under static conditions at 30 °C for up to 72 h. After incubation, the slides were gently removed, washed with 0.9% NaCl to eliminate non-adherent cells, and then fixed in 2.5% glutaraldehyde at 4 °C for 1 h. The fixed cells were sequentially dried in increasing concentrations of ethanol solutions (20% to 100%). Finally, the slides were gold-coated using sputter coating and examined under a FESEM (Carl Zeiss Auriga, Germany) for detailed structural analysis.

The biofilm formation ability of the isolate was also assessed quantitatively using the tissue culture plate (TCP) method. Bacterial cells in the log phase were diluted 1:100 in LB medium, and 100 µL of this suspension was dispensed into a 96-well polystyrene TCP in triplicate. The plate was then subsequently kept at 37 °C for 96 h. Following this, the wells were thoroughly washed with double-distilled water to remove any non-adherent cells. The remaining adherent cells were stained with 0.1% CV (crystal violet) (CV) for 20 min, followed by three washes with water, air drying, and a final rinse with 90% ethanol to dissolve the stain. The optical density (OD₆₀₀) of the biofilm was determined by ELISA reader (iMark™ Microplate Reader, Bio-Rad). Based

on the OD₆₀₀ values, the isolate was classified as a strong, moderate, or weak biofilm producer (Patra et al. 2024a).

Genomic sequencing, assembly, and functional analysis

The genomic DNA of the isolate underwent WGS analysis using the Illumina NovaSeq 6000 platform, conducted by Mediomix Diagnosis and Bioresearch (Bengaluru, India). Prior to sequencing, the purity and integrity of the DNA were evaluated using the Qubit™ dsDNA HS Assay Kit, analyzed by agarose gel electrophoresis (Thermo Fisher Scientific), and the quantification was done by Qubit 3.0 Fluorometer (Thermo Fisher Scientific). The preparation of the library involved the Enzymatic Fragmentation Kit 2.0 and Twist Universal Adapter System (Twist Bioscience, USA). A total of 300 ng of DNA was ligated with adapters at both ends, amplified for four PCR cycles using HiFi PCR Master Mix and primers, and purified with 1X AMPure XP beads (Beckman Coulter). The library was re-quantified with the Qubit 3.0 Fluorometer and Qubit dsDNA HS Assay Kit, and its quality was evaluated using the Agilent D1000 ScreenTape System on the 4150 TapeStation (Agilent), which analyzes DNA fragments between 35 and 1000 bp. The pooled libraries were sequenced, and the raw data underwent quality control checks using FastQC (v.0.11.5) and MultiQC (v.1.25). Low-quality reads and adapter sequences were removed with TrimGalore (v.0.6.5) (Fig. S1a, b, c, d). De novo assembly of the cleaned reads was performed using SPAdes (v.4.0.0) (Bankevich et al. 2012), and plasmid identification was carried out with PlasmidFinder 2.1. Scaffolds with fewer than 200 base pairs were excluded. The foreign contaminants present in the genome were analysed by NCBI FCS GX tool (<https://ncbiinsights.ncbi.nlm.nih.gov/2024/03/07/foreign-contamination-screen/>). Genome assembly metrics were generated using QUAST (v.5.0.2) (Gurevich et al. 2013), and the circular genome map was generated using the CGView server (Grant and Stothard 2008). The assembled genome was deposited in GenBank with the accession numbers BioProject PRJNA1058259 and BioSample SAMN39171102. Genome annotation was performed using the RAST server and Prokka pipeline (Aziz et al. 2008; Seemann 2014). The functional distribution of genes across different subsystems was analyzed using the RAST server, and the results were visualized in a pie chart created with Origin 2024b software. The functional categorization of proteins was conducted using the COG Classifier (v.1.0.5) (Deb and Das 2022), and the KEGG pathway for the entire genome of the isolate was predicted using the RAST server based on unique, core, and accessory genes.

Phylogenetic analysis, pangenome, and nucleotide identity comparisons

The 16S rRNA gene sequence of *Exiguobacterium* sp. E21L was extracted from its genomic data, and the phylogenetic analysis was performed. Phylogenetic relationships were inferred by aligning the 16S rRNA sequences of 50 reference strains sourced from the NCBI GenBank database. Sequence alignment was performed using Clustal Omega, and phylogenetic trees were constructed for the top 50 and top 10 alignment hits (Sievers and Higgins 2021). The tree was generated using the neighbor-joining method, with bootstrap values calculated over 1000 iterations to ensure the reliability of the branching structure (Felsenstein 1985). To further explore the genomic diversity, a pangenome analysis was conducted to identify the closest homologs based on the WGS analysis by using the Roary pan genome pipeline (v.3.11.2) incorporating 11 genomes from different *Exiguobacterium* species, including *Exiguobacterium sibiricum* 255–15, *Exiguobacterium chiriquhucha* RW-2, *Exiguobacterium undae* DSM 14481, *Exiguobacterium marinum* DSM 16307, *Exiguobacterium antarcticum* DSM 14480, *Exiguobacterium aurantiacum* DSM 6208, *Exiguobacterium oxidotolerans* JCM 12280, *Exiguobacterium indicum* HHS 31, *Exiguobacterium flavidum* strain HF60, *Exiguobacterium alkaliphilum* strain B-3531D, and *Exiguobacterium acetylicum* strain SI17. The average nucleotide identity (ANI) values between *Exiguobacterium* sp. E21L and the 11 reference genomes were calculated to assess genomic relatedness. Genomes with the highest ANI and fragment mapping values were identified and represented in a chord diagram. To further investigate the orthologous gene clusters, the four strains most closely related to our isolate were analyzed using the OrthoVenn2 web server (Xu et al. 2019), which compares clusters of orthologous genes based on gene ontology.

Assessment of antimicrobial resistance, heavy metal resistance, virulence factors, and genomic islands

ARGs in the genome of *Exiguobacterium* sp. E21L were detected through the CARD (Comprehensive Antibiotic Resistance Database) (Alcock et al. 2020), applying both stringent and relaxed identity thresholds. The whole genome of *Exiguobacterium* sp. E21L was then queried against BacMet (Antibacterial Biocide and Metal Resistance Genes Database) experimentally confirmed database to validate further and identify ARGs and HMRGs (Pal et al. 2014). The distribution of these ARGs, categorized by their mechanisms of action, was visualized in a pie chart created with Origin2024b. Additionally, functional annotations from the KEGG database were incorporated into the analysis to enhance the understanding of the genes' roles. Potential

virulence-associated genes were identified by performing BLASTp searches against the virulence factor database (VFDB) (Liu et al. 2019). Moreover, genomic islands (GIs), which are believed to be acquired via HGT and are potentially linked to AMR or pathogenicity, were identified using the IslandPath-DIMOB and SIGI-HMM methods integrated into the IslandViewer 4 tool (Bertelli et al. 2017).

Results

Isolation and characterization of the bacterial strain

The test isolate demonstrated vigorous growth on LB agar (M1151; Himedia) at both pH 7 and temperatures of 30 °C and 37 °C within 24 h (Fig. S2a, c). Morphological analysis revealed the presence of typical Gram-positive rod-shaped cells, with colonies ranging from 1–2 mm in diameter. The colonies displayed a mucoid, shiny, smooth, and moist appearance, with a light yellow colour and uniform growth patterns in regions of lower colony density on the LB agar. The metabolic profile of strain E21L indicated its ability to utilize various sugars, amino acids, and intermediates, including D-fucose, L-fucose, and D-serine. In contrast, it failed to metabolize D-melibiose, stachyose, and dextrin A2 (Table S2). Further examination using FESEM revealed short, rod-shaped cells approximately 1–2 µm long and 0.5 µm across (Fig. 1a). These cells aggregated into larger clusters, though without a consistent, uniform arrangement, showing sporadic patterns of aggregation (Fig. 1c). The isolate was identified as a member of the *Exiguobacterium* genus based on 16S rRNA gene sequencing, and the corresponding sequence has been submitted to GenBank with the accession number OQ913754. The phylogenetic analysis highlighted its close relationship to *Exiguobacterium mexicanum* strain IMTB-3094. Antibiotic susceptibility tests (AST) revealed that strain E21L was resistant to several antibiotic classes, including beta-lactams, fluoroquinolones, chloramphenicol, aminoglycosides, rifamycins, tetracyclines, sulfonamides, macrolides, and polypeptides while maintaining susceptibility to cephalosporins (Table 1). The MARI for strain E21L was calculated to be 0.88, indicating a high resistance level. The growth kinetics under heavy metal stress demonstrated that strain E21L exhibited enhanced growth at increasing concentrations of heavy metals, with the highest growth observed at 5 mM concentrations of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ and $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$, as shown in Fig. S3.

PCR-based detection of key ARGs, MGEs and HMRGs

Beyond AST, the molecular characteristics of strain E21L were further investigated using PCR amplification. A total of nine ARGs, including *blaNDM*, *qnrS*, *ermF*, *Sul1*, *blaGES*,

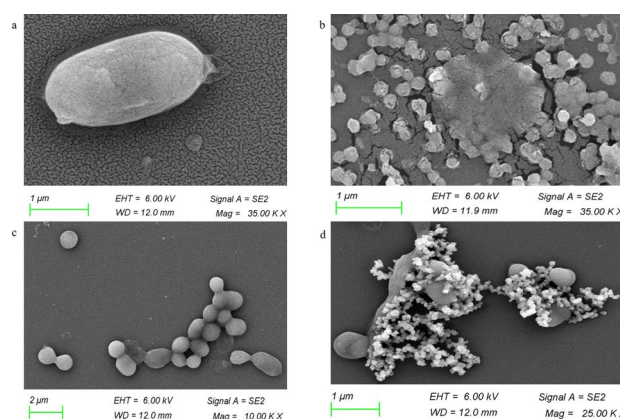


Fig. 1 **a** Morphology of the cells of *Exiguobacterium* sp. E21L under FESEM, **b** morphology of the cells during biofilm formation, **c** interbacterial adherence has been observed without EPS formation in control set, **d** interbacterial adherence has been observed with the formation of EPS under antibiotic concentration vancomycin ($30 \mu\text{g ml}^{-1}$)

tetW, *blaTEM*, *tetQ* and two MGEs, namely *IS26* and *intl1*, were successfully amplified from a panel of 13 ARGs and 2 MGEs. Additionally, the strain was found to harbor three HMRGs, *cZcD*, *nikA*, and *CopB*, which tested positive from a panel of six HMRGs. A Gel electrophoresis plots of the PCR amplification products are provided in Fig. S4. These findings underscore a notable level of both, antibiotic and heavy metal resistance within this strain.

Auto-aggregation, cell hydrophobicity, and static mono-compartmental assays

The hydrophobicity tests revealed that *E21L* exhibited hydrophobicity levels greater than 80% when exposed to several organic solvents, including hexane, hexadecane, ethyl acetate, and xylene (Fig. S5a). In the auto-aggregation assay, the strain showed initial adhesion at 78.51% after 24 h, which progressively increased to approximately 90% at 48 h, along with the formation of visible flocs, indicating enhanced aggregation over time (Fig. S2d, e). In the static mono-compartmental assay, E21L demonstrated significant survival under simulated GI conditions, with colony-forming units (CFU mL^{-1}) increasing throughout the experiment, maintaining similar cell densities ($p > 0.05$). A comparison of log CFU mL^{-1} at 0 and 360 min highlighted considerable bacterial growth ($p > 0.05$), suggesting effective adaptability in this environment (Fig. S5b).

In Vitro pathogenesis and structural insights of *Exiguobacterium* sp. E21L

The biofilm-forming ability of *Exiguobacterium* sp. E21L was quantitatively assessed via the TCP biofilm assay.

Table 1 Antibigram profiling of *Exiguobacterium* sp. E21L

Drug classes	Antibiotics	Concentration	Inhibition zone diameter (mm) ^a	Susceptibility ^b
Beta lactams	Penicillin-G	10U	14	R
	Ampicillin	10 µg	4	R
	Imipenem	10 µg	17	R
Sulfonamide	Cotrimoxazole	30 µg	0	R
Fluroquinolone	Ciprofloxacin	5 µg	0	R
	Ofloxacin	5 µg	0	R
	Norfloxacin	10 µg	12	R
Chloramphenicol	Chloramphenicol	30 µg	0	R
Aminoglycosides	Streptomycin	10 µg	0	R
	Gentamycin	10 µg	0	R
	Kanamycin	30 µg	8	R
Rifamycin	Rifampicin	5 µg	0	R
Tetracycline	Tetracycline	30 µg	14	R
Cephalosporine	Cefoxitin	30 µg	20	S
Macrolides	Erythromycin	15 µg	18	R
	Azithromycin	15 µg	16	I
Polypeptide	Vancomycin	5 µg	12	R

^aValues are the average of three repeated experiments^bThe antibiotic susceptibility of the strain was classified into S-Susceptible, R-Resistant

Analysis of triplicate technical readings from two biological replicates resulted in an average OD₆₀₀ of 1.879 for biological sample 1 and 2.235 for biological sample 2, ranging between 1.462 and 2.594. These findings suggest that *Exiguobacterium* sp. E21L has a moderate biofilm formation capacity, implicating its potential pathogenicity (Fig. S2f). In the motility assay, E21L exhibited active swarming motility from the centre toward the edges of the plate within 24 h of incubation, indicating a notable motility trait (Fig. S2b). FESEM imaging revealed increased secretion of extracellular polysaccharides (EPS) between colonies when exposed to subinhibitory vancomycin concentrations, in contrast to untreated cells (Fig. 1c, d). Additionally, bacterial adherence was observed in both sessile biofilm and planktonic states, as illustrated by the accompanying images (Fig. 1b, c).

Subsystems, orthologous gene clusters, and gene ontology-based analyses

The genome of *Exiguobacterium* sp. E21L was sequenced to a length of 3,062,037 bp with a high average coverage depth of 903X and GC content of 53.3% (Fig. 2) (Table S4). The contamination analysis revealed the presence of 53 contaminant sequences comprising of a small genome fraction of 20,749 bp (Table S5). Subsequent analysis revealed the absence of plasmids, indicating that the strain contains a single circular chromosome composed of 1050 contigs. The genome comprises 3894 protein-coding genes and 73

RNA genes. The L50 and N50 values were found to be 2 and 627,591, respectively. Genome annotation followed by subsystem analysis identified 25 major subsystems, with amino acid derivatives, carbohydrates, and protein metabolism being the most prominent, followed by subsystems related to nucleosides, nucleotides, RNA metabolism, and DNA metabolism (Fig. 3a). Functional annotation of the protein-coding genes was performed using the clusters of orthologous genes (COG) classifier, which categorized 2388 genes into various functional groups (Fig. 3b). Notably, 224 genes were associated with the biogenesis cluster, ribosomal structure, and translation, while the carbohydrate transport and metabolism cluster contained 202 genes. The amino acid metabolism and transport cluster comprised 189 genes, and 252 genes were linked to cell membrane, cell wall, or envelope biogenesis. Furthermore, 165 genes were involved in transcription, 147 in signal transduction mechanisms, and 62 were related to defence mechanisms. The mobilome category contained 4 genes related to prophages and transposons. Among the defence mechanisms, several genes were associated with AMR, such as those encoding the multidrug export protein *MepA*, the multidrug resistance ABC transporter, the high-affinity zinc uptake system ATP-binding protein *ZnuC*, *ZnuB*, the Zinc-specific metallo-regulatory protein, and the sec translocation accessory complex subunit *YajC*. Other resistance-related genes included multidrug resistance efflux pumps (e.g., *BcrA*, *EbrB*, and *EmrY*), antibiotic-modifying enzymes, ABC-type antimicrobial peptide

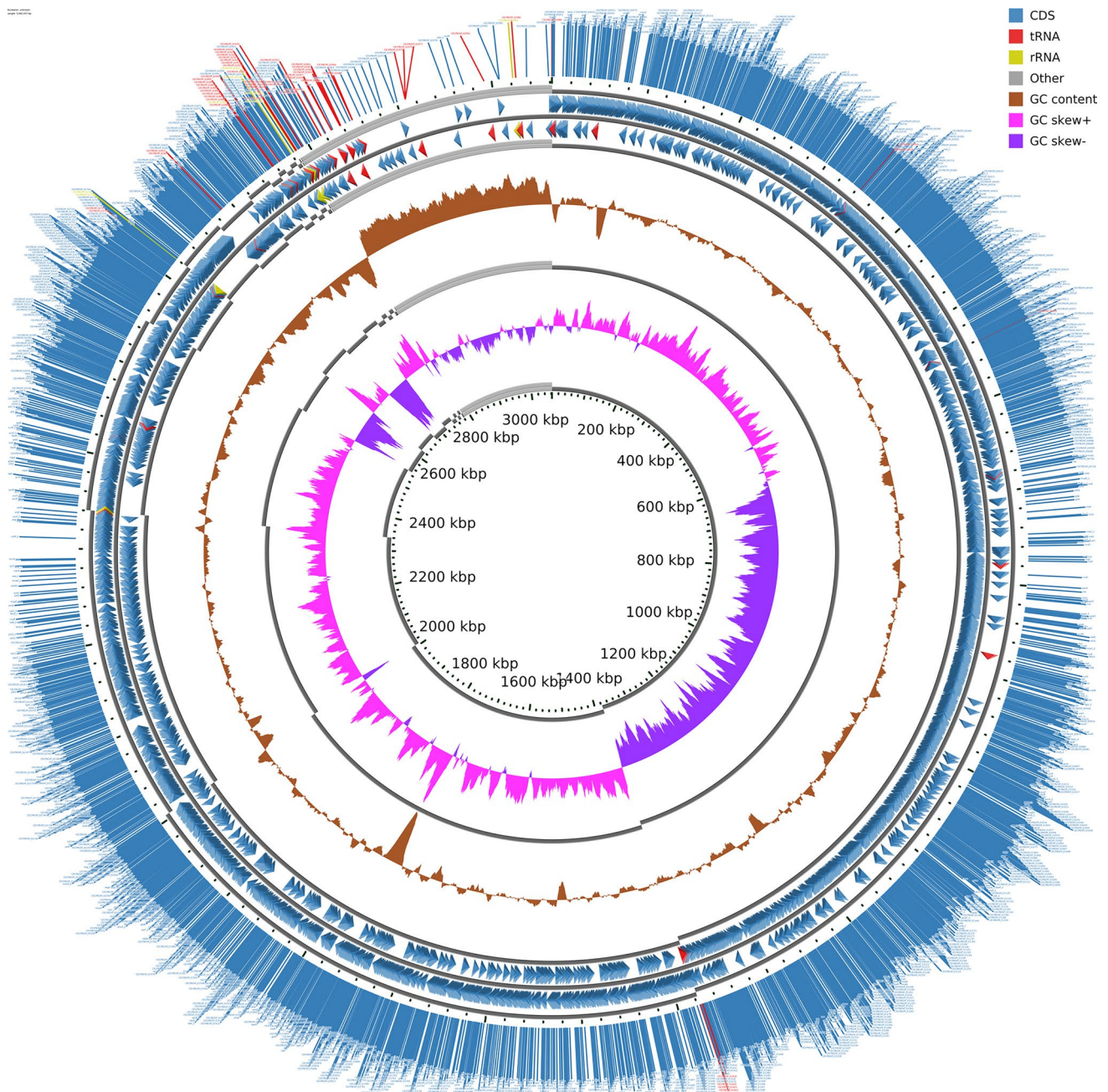


Fig. 2 Genomic map of *Exiguobacterium* sp. E21L displaying the distribution of coding sequences (CDS), transfer RNA (tRNA), ribosomal RNA (rRNA), and GC content over the genome. The map

also includes annotated genes and regions showing ARGs, HMRGs, MGEs, and virulence factors

transport systems, and ABC-type bacteriocin/lantibiotic exporters. These findings highlight the extensive repertoire of resistance mechanisms in the *Exiguobacterium* sp. E21L genome, reinforcing its potential for AMR.

Phylogenetic positioning and genome similarity analysis

Phylogenetic analysis based on 16S rRNA sequence of *Exiguobacterium* sp. E21L revealed that *Exiguobacterium* sp. E21L shares a close evolutionary relationship with *Exiguobacterium mexicanum* strain IMTB-3094 (Fig. 3c, d). Further pangenome analysis indicated that *Exiguobacterium*

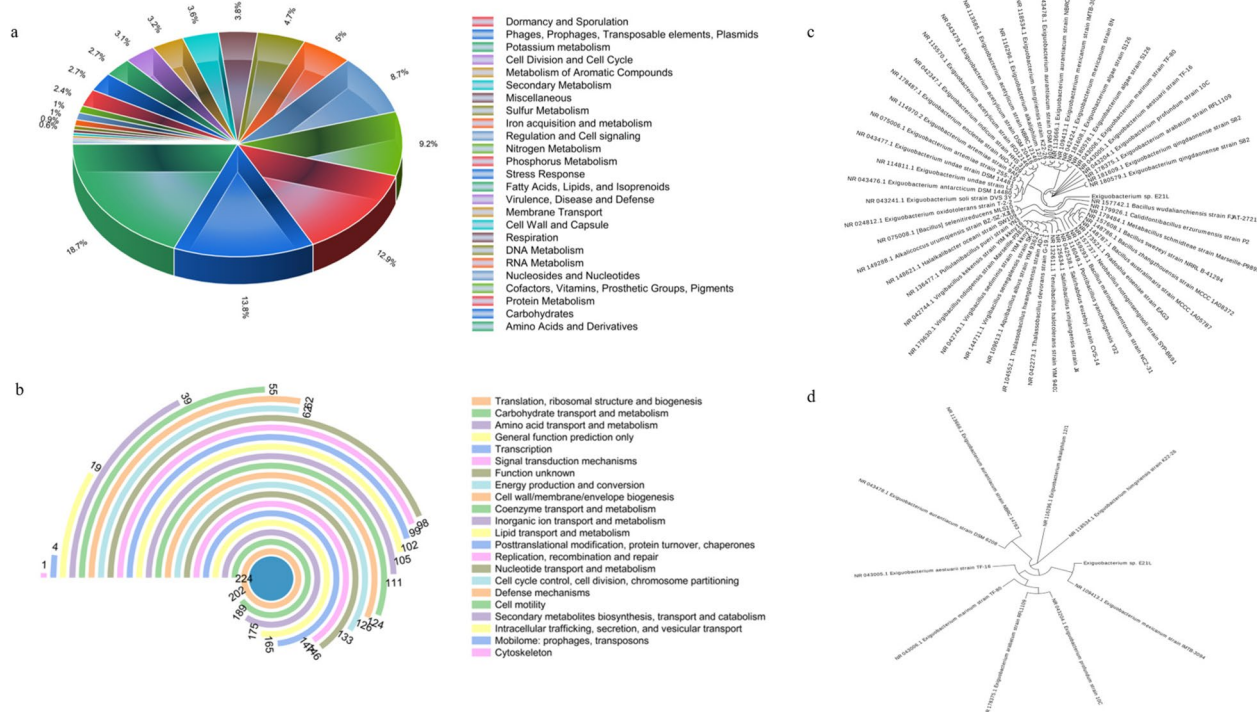


Fig. 3 **a** Pie chart showing genes distribution among different in *Exiguobacterium* sp. E21L, **b** COG analysis showing different clusters of orthologous groups in *Exiguobacterium* sp. E21L, **c** a phylogenetic tree based on 16S rRNA GENE illustrating the evolutionary relationship of *Exiguobacterium* sp. E21L with 50 other *Exiguobac-*

terium strains from clinical and environmental sources. The numbers at the nodes indicate bootstrap values based on 1000 iterations, **d** 16 s rRNA based phylogenetic tree of the closely related top 10 species

chiriquhucha RW-2 is the most closely related species to E21L (Fig. 4a). Therefore, the ANI and fragment mapping were performed with the four closest homologs to confirm this relationship further. The highest ANI value was observed between *Exiguobacterium* sp. E21L and *Exiguobacterium chiriquhucha* RW-2, with a value of 95.58% and 870 aligned fragments, confirming *E. chiriquhucha* as the closest related species (Fig. 4b).

Orthologous groups and KEGG pathway distribution

The parSNP phylogenetic tree analysis using multiple sequence alignment and the Roary matrix corroborated *Exiguobacterium chiriquhucha* RW-2 as the closest relative to *Exiguobacterium* sp. E21L (Fig. 4c). A further gene ontology-based comparison of gene clusters across the genomes of four closely related strains and *Exiguobacterium* sp. E21L identified 2589 gene clusters in *Exiguobacterium marinum* DSM 16307, 2889 in *Exiguobacterium chiriquhucha* RW-2, 2802 in *Exiguobacterium alkaliphilum* B-3531D, 2830 in *Exiguobacterium aurantiacum* DSM 6208, and 2780 clusters in *Exiguobacterium* sp. E21L. A common set of 2283 gene clusters was shared across all five genomes, with 282

clusters present in four, out of the five genomes (Fig. 5a–c). Among the two closely related strains, *Exiguobacterium chiriquhucha* RW-2 and *Exiguobacterium* sp. E21L, 2716 gene clusters were found to be conserved, further supporting their genomic similarity (Fig. 5d). To investigate species-level divergence, expansion, and contraction analysis was performed across these five genomes, offering insights into their adaptive mechanisms and evolutionary trajectories (Fig. S6). Additionally, KEGG pathway analysis highlighted the presence of genes involved in xenobiotic biodegradation and metabolism, including a beta-lactam resistance pathway, suggesting the potential for AMR in *Exiguobacterium* sp. E21L. A comprehensive study of the metabolomic network of this isolate was also conducted (Fig. S7).

Identification of genes associated with resistance and virulence

The analysis of AMR genes in *Exiguobacterium* sp. E21L revealed four critical genes through AMRfinder and BlastP analyses, using stringent criteria and the CARD database. These genes included the multidrug resistance *EbrB*, which confers resistance via an antibiotic efflux mechanism, and

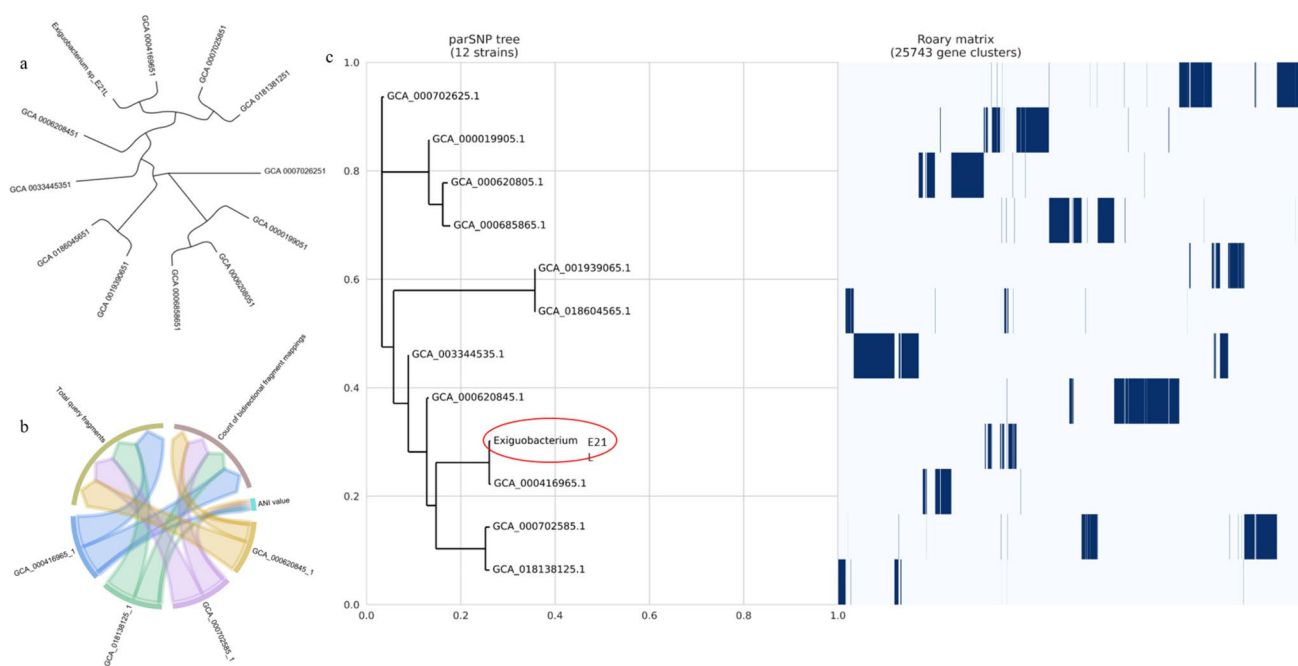


Fig. 4 Pan-genome analysis of *Exiguobacterium* sp. E21L and 11 closely related *Exiguobacterium* strains **a** phylogenetic tree based on pan-genome analysis, **b** the average nucleotide identity analysis between

E21L genome and 4 most related reference genomes, **c** parSNP based tree with Roary matrix-based on pangenome analysis, where the presence (blue) or absence (white) of core and accessory genes is indicated

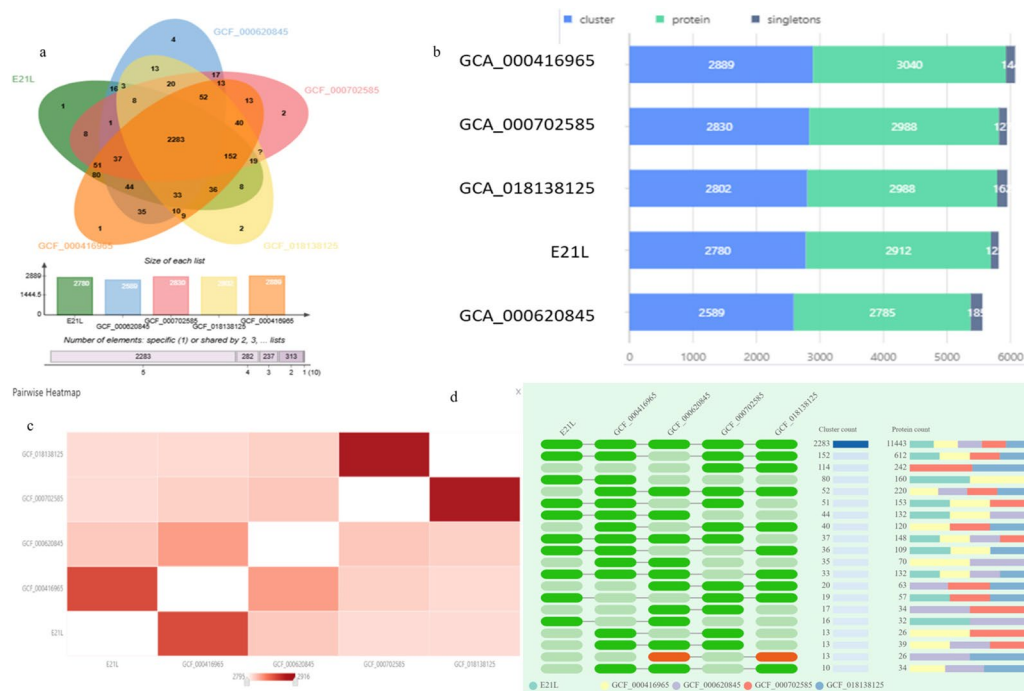


Fig. 5 Comparative genome analysis, **a** Venn diagram and bar chart illustrating the unique and shared orthologous gene clusters and their counts among closely related strains *Exiguobacterium marinum* strain DSM 16307 (GCF_000620845), *Exiguobacterium aurantiacum* strain DSM 6208 (GCF_000702585), *Exiguobacterium alkaliphilum* strain B-3531D (GCF_018138125), and *Exiguobacterium chiriquicha* strain RW-2 (GCF_000416965). **b** stacked bar plot comparing differences in

gene clusters, total proteins, and singletons, **c** pairwise heatmap highlighting similar strains based on overlapping gene clusters. **d** heat map showing the distribution of clusters based on orthologous genes of E21L with other 4 closely related species from OrthoVenn2 analysis and stacked bar plot showing the number of proteins comprised in clusters of respective strains

gene from the *vanA*, *vanM*, and *vanG* clusters, namely *vanW* (*vanG*), which contribute to vancomycin and teicoplanin resistance through alterations in antibiotic targets (Table S6). When the criteria was relaxed, the analysis identified a total of 273 AMR genes, of which 145 exhibited bit scores of 100 or higher (Fig. 6a) (Table S7). To further validate the AMR potential of these genes, BLAST analysis of these genes against BacMet database was performed and 103 genes out of the 273 genes identified based on relaxed criteria, achieved high bit score of greater than 100, proving high antimicrobial potential of the *Exiguobacterium* sp. E21L isolate (Table S7). Diverse arrays of genes responsible for antibiotic resistance were detected, such as erythromycin, bacitracin, acriflavine, tetracycline, ciprofloxacin, chloramphenicol, and others. These genes employ multiple mechanisms, including antibiotic efflux, target alteration, beta-lactamase-mediated inactivation, target protection, target replacement, and reduced permeability (Fig. 6c). The resistome analysis aligned with the phenotypic data confirmed the presence of multiple ARGs associated with resistance to a wide range of antibiotics (Table 1). Notably, the E21L genome harbored a substantial pool of multidrug resistance genes, such as *bcrA*, *MepA*, *YheI*, *YheH*, *BmrA*, *MarR*, *MsmX*, *DrrA*, *DrrC*, *EbrB*, *NorM*, *Mdr3*, *EmrY*, and

YwqF, among others (Tables S7 and S8). Additionally, several SNP mutations were identified in key genes, including *rpoB*, *gryA*, *gyrB* EF-Tu, *murA*, *thyA*, *PBP2x*, *fabI*, *pbp1*, *liaS*, *liaF*, *fusA*, *frxA*, *ndh* and *cls*, further supporting the strain's multidrug resistance profile.

Several virulence factors were also detected in the E21L genome. These included genes involved in chemotaxis (e.g., *McpU*, *CheV*, *CheW*, *CheA*, *CheY*, *PomA*), chemotactic enzymes (e.g., Chemotaxis protein methyltransferase), motility related genes (e.g., *SwrC/D*, *MotB*, *FlhE/G/P/Q/S/M/E/W*, *FlgC/B/G*, *FlhA/B/F*, *YlxH*, Flagellin), quorum sensing and biofilm formation (e.g., *BdlA*, *LuxS*, *DnaK*, *LsrA*, *SasA*, *EpsH/M*, *PHGDH*, *MTA*), and conserved virulence factors (e.g., *cvfB*), which contribute to the pathogenic potential of the strain. Further genomic exploration revealed a robust heavy metal resistance profile consistent with the phenotypic tolerance observed for various heavy metals. Genes involved in resistance to cadmium, zinc, cobalt copper, iron and arsenite were identified, including the cadmium, cobalt, and zinc/H(+)–K(+) exchanger, cadmium, zinc, and cobalt pumping ATPase, zinc-transporting ATPase, cadmium resistance transcriptional regulatory protein *CadC*, high-affinity zinc uptake system ATP-binding protein *ZnuC*, high-affinity zinc uptake system membrane

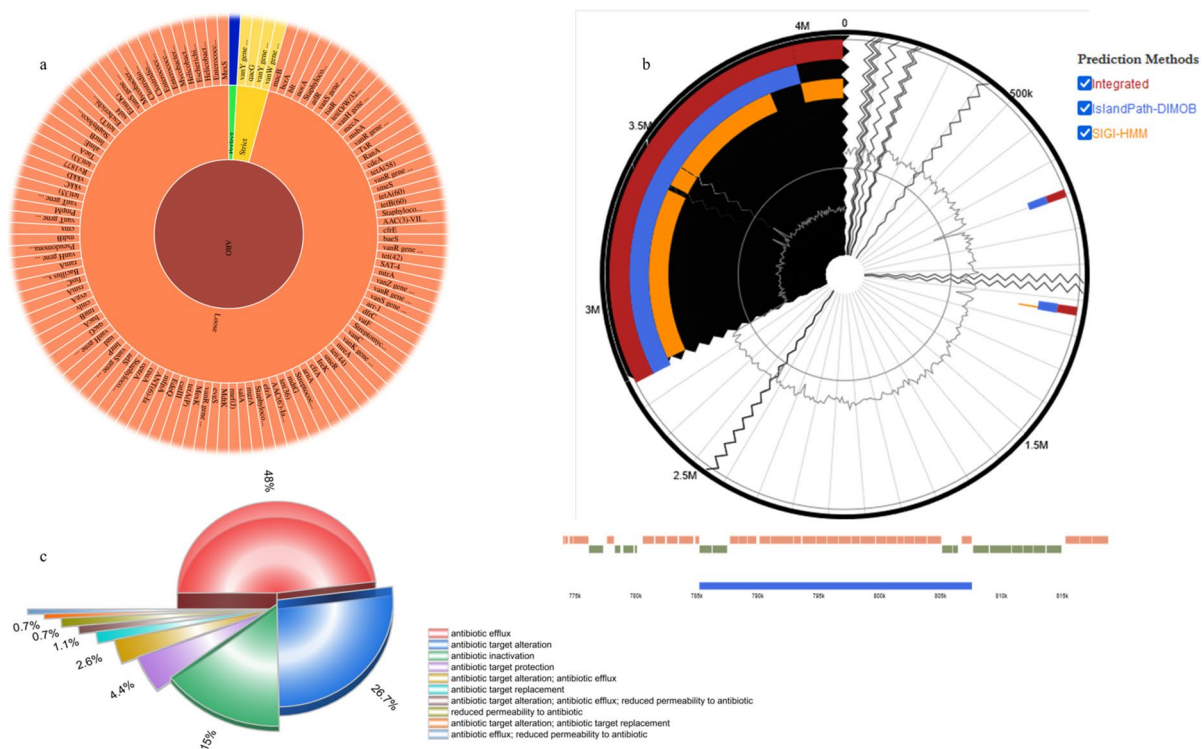


Fig. 6 **a** ARGs detected in the resistome of *Exiguobacterium* sp. E21L. **b** circular diagram illustrating the predicted GIs in *Exiguobacterium* sp. E21L genome coloured based on the prediction methods, red indicates an integrated analysis, blue represents IslandPath-

DIMOB prediction, orange represents SIGI-HMM prediction. **c** different AMR mechanisms mediated by proportions of ARGs in *Exiguobacterium* sp. E21L

protein *ZnuB*, zinc transporter *ZupT*, cobalt–nickel resistance proteins (*CnrA*), multicopper oxidase *mco*, copper-exporting P-type ATPase B, copper homeostasis protein *CutC*, iron-uptake system permease protein *FeuC*, putative iron export permease protein *FetB*, putative chromate transport protein, iron uptake protein A1, arsenical pump-driving ATPase and arsenical pump membrane protein (Tables S7 and S8). Additionally, genes encoding proteins that utilize heavy metals as cofactors for catalytic processes were identified, further reinforcing the strain's adaptability to heavy metal stress.

Mobilome and genomic islands in *Exiguobacterium* sp. E21L

The genome of *Exiguobacterium* sp. E21L revealed the presence of several important insertion sequences and transposons, including members of the *IS1595* family, such as *ISSpg11*, *ISCac2*, *IS200/IS605* family transposases *ISTel3*, and *Tn3* family transposase *ISAau4* (Fig. 6b). Key genes associated with various metabolic processes were identified within the GIs, including rhamnose metabolism (*rhaR*), tRNA-specific methylthio-transferase (*mnmA*), caseinolytic protease (*clpC*), and sulfur transfer factor (*sufE*) (Table S3). Additionally, 56 hypothetical proteins were detected, which are likely part of horizontally transferred genomic regions and may contribute to the bacterium's adaptability and pathogenic potential. Overall, 82 hypothetical proteins and 41 known genes were found in the GIs of *Exiguobacterium* sp. E21L (Table S3). Notably, several genes related to AMR and stress adaptation were identified, such as *ugd*, *rfbA*, *hldD*, *uvrD*, *recF*, *gyrA*, *gyrB*, *glmU*, *xerD*, and *nimR*. These genes are implicated in polysaccharide biosynthesis, DNA repair, RNA metabolism, and AMR mechanisms. The presence of these AMR genes within the GIs indicates towards HGT of these genes, suggesting that these regions may play a role in the bacterium's ability to adapt to environments rich in antimicrobial compounds.

Discussion

Antibiotics and heavy metals resistance in *Exiguobacterium* sp. E21L

Exiguobacterium sp. E21L offers compelling insights into the adaptation of endophytes in contaminated environments. It demonstrates dual resistance to antibiotics and heavy metals, indicating a potential role in enhancing plant resilience to environmental stressors. Notably, the strain remained susceptible to cephalosporins, suggesting variability in resistance mechanisms across beta-lactam

antibiotics. Similar studies on endophytes from polluted habitats have reported broad-spectrum resistance, highlighting the adaptive advantage conferred by endophytic colonization in such settings (Muhammad et al. 2024). The concomitant occurrence of antibiotic and metal resistance in E21L, combined with its enhanced growth under elevated copper and zinc concentrations, aligns with findings on resistance co-selection via efflux pumps and metal-binding proteins (Mokni-Tlili et al. 2023; Prabhu et al. 2024). This suggests that E21L may mitigate metal toxicity and reduce oxidative stress, potentially aiding plant survival in polluted soils (Xiao et al. 2024). The high MARI value and observed co-selection of resistance traits point to a possible role for E21L in the HGT of resistance genes within plant microbiomes, a phenomenon commonly reported in endophytes from urban agricultural systems (Mahdi et al. 2022). This is particularly significant in urban farming, where sewage sludge amendments may inadvertently promote the persistence of resistant strains in crops, posing a risk of AMR dissemination. The close phylogenetic relationship of E21L to other members of the *Exiguobacterium* genus, known for thriving in extreme environments, further supports its potential as a plant-associated bacterium capable of promoting plant growth and stress tolerance in contaminated soils (Pandey 2020; Zhang et al. 2021). Additionally, E21L may serve as a bioremediator, mitigating environmental contamination while facilitating plant adaptation, adding ecological value to its role in polluted agroecosystems (Sur and Sathivelu 2024).

Capability for gastrointestinal persistence and colonization

Exiguobacterium sp. E21L demonstrates a notable ability to survive and adapt under simulated GI conditions, shedding light on its potential for gut colonization. High hydrophobicity levels (> 80%) in the presence of various organic solvents (hexane, hexadecane, ethyl acetate, xylene) indicate a strong cell surface affinity, likely enhancing adhesion to epithelial surfaces, a critical factor for colonization (Nataraj et al. 2021). Hydrophobicity is a well-documented trait linked to bacterial adherence and initial colonization in the GI tract (Han et al. 2021; Zheng et al. 2021). The strain's significant auto-aggregation capacity supports its ability to form microcolonies and withstand shear forces in the gut, a key aspect of persistent colonization (Pu and Rowe-Magnus 2018; Béchon and Ghigo 2022). In a static mono-compartmental model assay, E21L exhibited increased viable cell counts over time, demonstrating its resilience to harsh GI conditions, including low pH, bile salts, and digestive enzymes.

Comparable results in probiotic studies link such survival traits to enhanced colonization potential (Dias et al. 2019). Notably, improved EPS production under stress conditions further supports the strain's persistence and adaptability (Gong et al. 2024). The ability to maintain viable cell densities suggests that E21L could transiently or stably integrate into the gut microbiota under favourable conditions. Moreover, ARGs in such strains raise concerns about AMR dissemination in the gut. Recent studies highlight that probiotics harbouring ARGs may tolerate subinhibitory antibiotic doses, facilitating adaptive evolution and increasing AMR transmission risk in the gut environment (Tian et al. 2024).

Motility, biofilm formation, and phenotypic traits

The biofilm formation and motility of *Exiguobacterium* sp. E21L underscore its environmental resilience and potential clinical relevance. Biofilm production, especially when coupled with the secretion of EPS under subinhibitory concentrations of vancomycin, a last-resort antibiotic-points to a survival strategy that enhances antimicrobial tolerance and potentially aids in HGT, thereby compounding AMR risks (Vani et al. 2023; Ren et al. 2024). This adaptability is concerning, given the strain's ability to persist in hostile environments, including those with heavy metal contamination and antibiotic pressure. The dual adhesion capability in planktonic and sessile states further mirrors a hallmark of opportunistic pathogens, increasing its potential for surface colonization and persistence in clinical and environmental niches (Shree et al. 2023). While the clinical pathogenicity of *Exiguobacterium* species remains underexplored, these findings highlight a need for further genetic and phenotypic investigations. Specifically, elucidating the molecular mechanisms behind biofilm formation and motility could clarify how this strain adapts to antimicrobial and environmental stressors, as comparable studies in related extremophilic bacteria have demonstrated critical regulatory genes governing these traits (Pavez et al. 2023).

Genomic insights and functional adaptations of *Exiguobacterium* sp. E21L

The genome of *Exiguobacterium* sp. E21L offers a comprehensive view of its genetic and functional potential and exhibits a versatile genetic architecture supporting diverse metabolic capabilities. Identifying genes linked to AMR, including efflux pumps, ABC transporters, and metal ion transporters, underscores its potential role in resistance to various antimicrobial agents. Multidrug resistance efflux pumps such as *BcrA*, *EbrB*, and *EmrY*, commonly associated

with efflux-mediated resistance in bacteria from sludge-amended soils, were prominent in E21L (Markowicz et al. 2021). Additionally, the high-affinity zinc uptake system *ZnuC*, a known virulence-associated factor, suggests that metal stress could drive the co-selection of AMR traits (Singh et al. 2024a, b). The discovery of prophage- and transposon-related genes indicates potential HGT events, enhancing the strain's ability to acquire and disseminate resistance genes within microbial communities. Functional annotations revealed genes associated with carbohydrate metabolism, amino acid transport, and nucleic acid metabolism, highlighting its adaptability to nutrient-variable environments, where comparative genomic analyses further confirm that *Exiguobacterium* species possess intrinsic resistance mechanisms against heavy metals and antibiotics, supporting their survival in contaminated settings. Therefore, the genomic features of E21L suggest its potential as a model for studying resistance mechanisms in bacteria exposed to anthropogenic pressures. The diverse AMR determinants in an endophytic bacterium underscore its potential role in the plant-associated resistome, with implications for AMR transmission in agroecosystems. Further transcriptomic and proteomic studies under varying stress conditions are required to elucidate further the regulatory pathways underlying these resistance traits and their environmental significance.

Comparative genomics, phylogenetic insights, and functional genomic features

The phylogenetic analysis of *Exiguobacterium* sp. E21L reveals its evolutionary proximity to *Exiguobacterium chiriquicha* RW-2, indicating species-level relatedness and core genome conservation. This result also aligns with earlier reports on *Exiguobacterium* species, recognized for their genomic plasticity and resilience in extreme and contaminated environments (Shen et al. 2024). The close association with *E. mexicanum* and other extremotolerant species underscores its ecological versatility. The overlap in conserved gene clusters with *E. chiriquicha* RW-2 suggests shared adaptive traits crucial for survival in metal-contaminated habitats, aligning with findings on *Exiguobacterium*'s persistence under environmental stress (Zhang et al. 2021). KEGG pathway analysis further supports this adaptability, highlighting genes involved in beta-lactam resistance and AMR pathways, likely driven by co-selection under anthropogenic pressures. Additionally, variations in gene clusters related to transport, signalling, and defence mechanisms point to evolutionary adaptations shaped by environmental stressors (Parrilli et al. 2021). These features emphasize the dual ecological and biotechnological potential of *Exiguobacterium*

sp. E21L, particularly in bioremediation, given its xenobiotic degradation pathways.

Insights into resistance and virulence potential of *Exiguobacterium* sp. E21L

The resistome analysis of *Exiguobacterium* sp. E21L revealed a complex network of ARGs, including multiple efflux pumps, regulatory elements, and specific drug resistance determinants. Among these, efflux pumps play a crucial role in bacterial resistance, actively exporting toxic compounds and reducing intracellular antibiotic accumulation. Overexpression of efflux pumps is a well-established driver of MDR in bacteria, contributing to reduced antibiotic susceptibility and complicating treatment options (Seukep et al. 2022). The detection of *MepA*, a gene frequently identified in hospital wastewater (Zulkeflee et al. 2019), household garbage (Liang et al. 2020), and lettuce-soil systems exposed to antibiotics (Shen et al. 2021), highlights the potential for environmental dissemination. Similarly, *EmrY* has been reported in wastewater river continua (Read et al. 2024), as well as in agricultural and urban-influenced watersheds (Uyaguari-Díaz et al. 2017). Notably, the expression of *EmrY* has also been shown to increase upon zinc exposure, suggesting potential co-selection mechanisms between antibiotic and metal resistance (Rihacek et al. 2023). Efflux pumps such as *NorM* and *MepA* confer antibiotic resistance and may reduce bacterial susceptibility to disinfectants, further complicating infection control strategies. Beyond multidrug efflux systems, identifying vancomycin resistance gene clusters (*VanA*, *VanM*, *VanW*) is particularly concerning due to their clinical significance in glycopeptide resistance (Xu et al. 2010). These resistance determinants are often associated with Tn1546-type transposons, which, when carried on Inc18 and RepA_N conjugative plasmids, facilitate the horizontal transfer of *vanA* operons among clinical *Enterococcus* isolates. In contrast, *vanB* gene clusters are typically found within Tn1549-Tn5382 integrative conjugative elements (Wardal et al. 2023). These vancomycin resistance genes in an environmental isolate suggests their potential for dissemination beyond clinical settings, raising concerns about their mobility and persistence in anthropogenically impacted ecosystems. In addition to antibiotic resistance genes, the study also identified *bcrA* and *ugd*, which have been previously reported to co-localize with HMRGs, indicating the likelihood of co-selection in metal-contaminated environments (Ullmann et al. 2020; Kang et al. 2024). *bcrA* has been found on integrative conjugative elements such as *ICEcp1*, which is related to the Tn916 family, further emphasizing its potential for horizontal gene transfer. Similarly, *DrrA* and *DrrC*, known drug resistance determinants, have been

associated with MGEs like Tn6097, reinforcing their ability to disseminate across bacterial populations (Dziewit et al. 2012). The presence of recombination-associated genes like *xerD* within genomic islands in *E21L* further supports the likelihood of HGT, suggesting that these resistance genes could spread within environmental microbial communities, a trend that has been observed in previous studies (Li and Wang 2021). From a public health standpoint, the presence of these resistance determinants in an endophytic bacterium raises significant concerns. The detection of *gyrA* and *gyrB* mutations, which confer fluoroquinolone resistance (Shafiq et al. 2024), underscores the potential risk of MDR development in pathogenic strains. If these genes are mobilized into clinically relevant bacteria, they could significantly impact antibiotic efficacy, reducing treatment options for bacterial infections. The co-occurrence of ARGs and HMRGs within *E21L* is particularly worrisome in agriculture, where sludge-amended crops may act as reservoirs for resistant bacteria. The potential for these genes to enter the food chain necessitates stringent monitoring of AMR reservoirs in agroecosystems to mitigate the spread of resistance determinants. The presence of vancomycin resistance determinants alongside MDR-associated genes further underscores the growing concern over AMR proliferation in environmental settings.

The E21L's resistome encompassing genes for erythromycin, tetracycline, ciprofloxacin, and beta-lactams, correlates with AST results. However, discrepancies between phenotypic resistance and genomic data suggest the involvement of homologous or alternative genes, a phenomenon common in bacteria with complex resistomes (Baquero et al. 2021).

The detection of cadmium, zinc, and cobalt resistance operons, confirmed by PCR amplification of *cZcD*, *nika*, and *CopB* and supported by WGS data, underscores the strain's enhanced fitness in contaminated environments and its potential role in ARG propagation in plant-associated niches. Similar adaptive strategies have been observed in bacteria from metal-rich habitats (Gillieatt and Coleman 2024). While *czcABC* was not annotated in our study, the presence of cadmium, zinc, and cobalt-transporting ATPases, along with cadmium, cobalt, and zinc/H⁺-K⁺ antiporters, suggests that *E21L* possesses alternative metal resistance mechanisms that contribute to detoxification and homeostasis under heavy metal stress. For zinc resistance, the detection of a complete high-affinity zinc uptake system, including *ZnuC*, *ZnuB*, and *ZupT*, indicates efficient intracellular zinc regulation (Chanket et al. 2024). Copper resistance was supported by the detection of copper-exporting P-type ATPase B (*CopB*), and copper homeostasis protein *CuIC* highlighting metal efflux mechanisms. Although *copA* was not identified, previous studies indicate that *copB* alone can contribute to copper homeostasis by facilitating copper

export, thereby reducing intracellular toxicity (Adhikary et al. 2024). Iron resistance genes, including *FeuC*, *FetB*, and iron uptake protein A1, suggest active iron transport and homeostasis, while arsenical resistance determinants, such as the arsenical pump-driving ATPase and arsenical pump membrane protein indicate the ability to mitigate arsenic toxicity beyond direct metal resistance genes, several metalloregulation and stress-response genes were annotated, including, metallo-hydrolase (*YycJ*), ATP-dependent zinc metalloprotease (*FtsH*), metalloregulation DNA-binding stress protein (*mrgA*), and zinc-specific metallo-regulatory protein (*zur*). These genes play essential roles in bacterial adaptation to metal stress by regulating metal ion transport, detoxification pathways, and oxidative stress responses (Martino et al. 2002; Pinochet-Barros and Helmann 2018; Wu et al. 2024). While the annotated HMRGs in *E21L* may not form a complete resistance system, the combination of detected efflux transporters, high-affinity uptake systems, and regulatory elements suggests an effective metal resistance strategy.

Additionally, key SNP mutations in ARGs suggest that intrinsic mechanisms complement acquired ARGs, contributing to the robust resistance phenotype of E21L. The presence of diverse virulence factors in *Exiguobacterium* sp. E21L highlights its potential dual role as an environmental microorganism and an opportunistic pathogen. Genes associated with motility, chemotaxis, biofilm formation, and quorum sensing enhance bacterial adaptation and contribute to host interactions and potential pathogenicity. Chemotaxis-related genes such as *cheA*, *cheB*, *cheR*, *cheV*, and *cheY* are critical for adhesion and motility, playing a significant role in bacterial colonization of host tissues, as observed in *H. pylori* (Keilberg and Ottemann 2016). The upregulation of *pomA* of *A. pleuropneumoniae* in necrotic lung tissue further supports its relevance in host–pathogen interactions (Baltes and Gerlach 2004). Genes like *motB* and *CheW* have been linked to bacterial survival in heavy metal-contaminated environments, suggesting their involvement in environmental resilience (Dinesh et al. 2023). Previous studies have reported a higher prevalence of flagellar genes such as *flgC*, *flgG*, *fliG*, *fliM*, and *fliP* in uropathogenic *E. coli* and their role in host binding (*Salmonella enterica* Serovar Enteritidis) (Shippy et al. 2014). Similarly, *FlhA/B/F* detected in E21L are crucial for *Campylobacter jejuni* gut colonization in avian and mammalian hosts (Ren et al. 2018), indicating their broader ecological relevance. Furthermore, the conserved virulence factor *cyfB* plays a key role in pathogenicity, with mutations leading to reduced virulence in *S. aureus*, decreased hemolysin, DNase, and protease production (Matsumoto et al. 2007). The presence of *EpsH*, a known factor in bacterial-driven intestinal inflammation (Jones et al. 2014), and *Bdla*, which contributes to the acute virulence and chronic persistence of *Pseudomonas*

aeruginosa (Li et al. 2014), further strengthens the argument that E21L harbors factor that could influence host–pathogen interactions.

GIs in E21L, harbouring ARGs, hypothetical proteins, and metabolic genes, emphasize the role of HGT in shaping its genomic architecture. The presence of MGEs, including insertion sequences and transposases from the *IS1595* and *Tn3* families, aligns with previous studies highlighting their role in facilitating the mobility of ARGs (Yan et al. 2023). Notably, *PbpX*, a gene potentially linked to antibiotic resistance, was found adjacent to *ISSpgII*, suggesting its acquisition through HGT from environmental sources (Merlin et al. 2022). Furthermore, the presence of *ISAu4*, previously reported in MDR *Pseudomonas aeruginosa* from hospital wastewater, underscores its potential role in the mobilization of ARGs under selective pressure (Khan et al. 2022). The presence of *ISCac2*, also previously identified in *Lysinibacillus fusiformis*, a strain exhibiting crude petroleum oil degradation ability, further highlights the adaptability of these elements in diverse ecological niches (Diabankana et al. 2024). Despite detecting these transposons, our study did not identify direct ARG insertions within them. However, the proximity of these MGEs to genes implicated in AMR suggests a possible mechanism of cross-resistance via genetic rearrangements. Furthermore, integrating stress adaptation genes, such as those involved in polysaccharide biosynthesis and DNA repair, within GIs also strengthens their critical function in enhancing environmental resilience. Therefore, these findings highlight the need for continued surveillance of ARGs and HMRGs in plant-associated bacterial communities exposed to anthropogenic pollution.

Conclusion

Comprehensive WGS analysis of *Exiguobacterium* sp. E21L highlighted its potential as a vector for AMR transmission within plant endophytes, raising critical public health concerns. While *Exiguobacterium* sp. E21L has been previously recognized for its PGPR activity; our findings indicated its emerging role in the AMR landscape, particularly in environments impacted by sewage sludge application. This study underscores the pressing need for a One Health approach to mitigate AMR risks, aligning with Sustainable Development Goal 3. Additionally, it raises concerns about the inefficiency of existing anaerobic and aerobic sewage sludge treatment processes in effectively eliminating AMR determinants, thereby suggesting the adoption of advanced treatment technologies in STPs. The agricultural use of sewage sludge while beneficial for enhancing soil fertility, may inadvertently contribute to the proliferation of AMR, necessitating urgent policy interventions and promoting

safer agrarian practices. Moreover, the ability of *Exiguobacterium* sp. E21L to harbor several MGEs, coupled with its high motility, aggregation, and biofilm formation capabilities, underscores its ecological and clinical significance in AMR dissemination. Despite being isolated from control soil, the significantly lower MARI values in those isolates suggest that resistance development is strongly influenced by selective pressure in sludge-amended environments. This highlights the critical role of contaminated sludge in shaping plant-associated microbiota's resistome. Notably, the role of endophytes in contaminated plants as potential reservoirs of resistance determinants highlighted a crucial pathway for AMR transmission through the food chain, emphasizing the broader implications for human health and food safety. Therefore, the present findings call for immediate action to strengthen the regulatory frameworks and promote sustainable agricultural practices to minimize AMR risks associated with using treated sewage sludge.

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Author contributions M.P. Investigation, data curation, data analysis and writing-editing of the original draft. A.K.P. Data analysis, software, writing and editing. S.K.D. Conceptualization, supervision, funding acquisition, writing review and editing.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declared that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval Not applicable.

Informed consent Not applicable.

Consent for publication Not applicable.

References

- Abadía-García L, Cardador A, del Campo STM, Arvizu SM, Castaño-Tostado E, Regalado-González C, García-Almendarez B, Amaya-Llano SL (2013) Influence of probiotic strains added to cottage cheese on generation of potentially antioxidant peptides, anti-listerial activity, and survival of probiotic microorganisms in simulated gastrointestinal conditions. *Int Dairy J* 33:191–197. <https://doi.org/10.1016/j.idairyj.2013.04.005>
- Adhikary S, Saha J, Dutta P, Pal A (2024) Bacterial homeostasis and tolerance to potentially toxic metals and metalloids through diverse transporters: metal-specific insights. *Geomicrobiol J* 41:496–518. <https://doi.org/10.1080/01490451.2024.2340517>
- Alcock BP, Raphenya AR, Lau TT, Tsang KK, Bouchard M, Edalatmand A, Huynh W, Nguyen ALV, Cheng AA, Liu S, Min SY (2020) CARD 2020: antibiotic resistome surveillance with the comprehensive antibiotic resistance database. *Nucleic Acids Res* 48:D517–D525. <https://doi.org/10.1093/nar/gkz935>
- Arora PK, Sharma A, Mehta R, Shenoy BD, Srivastava A, Singh VP (2012) Metabolism of 4-chloro-2-nitrophenol in a gram-positive bacterium, *Exiguobacterium* sp. PMA. *Microb Cell Fact* 11:1–10. <https://doi.org/10.1186/1475-2859-11-150>
- Aslam B, Wang W, Arshad MI, Khurshid M, Muzammil S, Rasool MH, Nisar MA, Alvi RF, Aslam MA, Qamar MU, Salamat MKF (2018) Antibiotic resistance: a rundown of a global crisis. *Infect Drug Resist*. <https://doi.org/10.2147/IDR.S173867>
- Atkinson S, Chang CY, Sockett RE, Cámara M, Williams P (2006) Quorum sensing in *Yersinia enterocolitica* controls swimming and swarming motility. *J Bacteriol* 188:1451–1461. <https://doi.org/10.1128/jb.188.4.1451-1461.2006>
- Aziz RK, Bartels D, Best AA, DeJongh M, Disz T, Edwards RA, Formisano K, Gerdes S, Glass EM, Kubal M, Meyer F (2008) The RAST server: rapid annotations using subsystems technology. *BMC Genom* 9:1–15. <https://doi.org/10.1186/1471-2164-9-75>
- Baltes N, Gerlach GF (2004) Identification of genes transcribed by *Actinobacillus pleuropneumoniae* in necrotic porcine lung tissue by using selective capture of transcribed sequences. *Infect Immun* 72:6711–6716. <https://doi.org/10.1128/iai.72.11.6711-6716.2004>
- Bankevich A, Nurk S, Antipov D, Gurevich AA, Dvorkin M, Kulikov AS, Lesin VM, Nikolenko SI, Pham S, Prjibelski AD, Pyshkin AV (2012) SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. *J Comput Biol* 19:455–477. <https://doi.org/10.1089/cmb.2012.0021>
- Baquero F, Martínez JL, Lanza FV, Rodríguez-Beltrán J, Galán JC, San Millán A, Cantón R, Coque TM (2021) Evolutionary pathways and trajectories in antibiotic resistance. *Clin Microbiol Rev* 34:e00050. <https://doi.org/10.1128/CMR.00050-19>
- Béchon N, Ghigo JM (2022) Gut biofilms: bacteroides as model symbionts to study biofilm formation by intestinal anaerobes. *FEMS Microbiol Rev* 46:fuab054. <https://doi.org/10.1093/femsre/fuab054>
- Bertelli C, Laird MR, Williams KP, Simon Fraser University Research Computing Group, Lau BY, Hoad G, Winsor GL, Brinkman FS (2017) IslandViewer 4: expanded prediction of genomic islands for larger-scale datasets. *Nucleic Acids Res* 45:W30–W35. <https://doi.org/10.1093/nar/gkx343>
- Chanket W, Pipatthana M, Sangphukieo A, Harnvoravongchai P, Chankhamhaengdech S, Janvilisri T, Phanchana M (2024) The complete catalog of antimicrobial resistance secondary active transporters in *Clostridioides difficile*: evolution and drug resistance perspective. *Comput Struct Biotechnol J* 23:2358–2374. <https://doi.org/10.1016/j.csbj.2024.05.027>
- Chen P, Yu K, He Y (2023) The dynamics and transmission of antibiotic resistance associated with plant microbiomes. *Environ Int* 176:107986. <https://doi.org/10.1016/j.envint.2023.107986>
- Christaki E, Marcou M, Tofarides A (2020) Antimicrobial resistance in bacteria: mechanisms, evolution, and persistence. *J Mol Evol* 88:26–40. <https://doi.org/10.1007/s00239-019-09914-3>
- Collins MD, Lund BM, Farrow JAE, Schleifer KH (1983) Chemotaxonomic study of an alkalophilic bacterium, *Exiguobacterium*

- aurantiacum* gen. nov., sp. nov. Microbiology 129:2037–2042. <https://doi.org/10.1099/00221287-129-7-2037>
- da Costa WLO, Araújo CLDA, Dias LM, Pereira LCDS, Alves JTC, Araujo FA, Folador EL, Henriques I, Silva A, Folador ARC (2018) Functional annotation of hypothetical proteins from the *Exiguobacterium antarcticum* strain B7 reveals proteins involved in adaptation to extreme environments, including high arsenic resistance. PLoS ONE 13:e0198965. <https://doi.org/10.1371/journal.pone.0198965>
- Deb S, Das SK (2022) Phylogenomic analysis of metagenome-assembled genomes deciphered novel acetogenic nitrogen-fixing Bathyarchaeota from hot spring sediments. Microbiol Spectrum 10:e00352–e00422. <https://doi.org/10.1128/spectrum.00352-22>
- Diabankana RGC, Zhamalbekova AA, Shakirova AE, Vasiuk VI, Filimonova MN, Validov SZ, Safin RI, Afordanyi DM (2024) Genomic insights of wheat root-associated *Lysinibacillus fusiformis* reveal its related functional traits for bioremediation of soil contaminated with petroleum products. Microorganisms 12:2377. <https://doi.org/10.3390/microorganisms12112377>
- Dias C, Ribeiro M, Correia-Branco A, Domínguez-Perles R, Martel F, Saavedra MJ, Simões M (2019) Virulence, attachment and invasion of Caco-2 cells by multidrug-resistant bacteria isolated from wild animals. Microb Pathog 128:230–235. <https://doi.org/10.1016/j.micpath.2019.01.011>
- Dinesh R, Sreena CP, Sheeja TE, Charles S, Srinivasan V, Sajith V, Subila KP, Hariha P (2023) Metagenomics indicates abundance of biofilm related genes and horizontal transfer of multidrug resistant genes among bacterial communities in nano zinc oxide polluted soil. Sci Total Environ 859:160032. <https://doi.org/10.1016/j.scitotenv.2022.160032>
- Djordjevic SP, Jarocki VM, Seemann T, Cummins ML, Watt AE, Drigo B, Wyrsh ER, Reid CJ, Donner E, Howden BP (2024) Genomic surveillance for antimicrobial resistance—a one health perspective. Nat Rev Genet 25:142–157. <https://doi.org/10.1038/s41576-023-00649-y>
- Dziewit L, Baj J, Szuplewska M, Maj A, Tabin M, Czyzkowska A, Skrzypczyk G, Adamczuk M, Sitarek T, Stawinski P, Tudek A (2012) Insights into the transposable mobilome of *Paracoccus* spp. (Alphaproteobacteria). PLoS ONE 7:e32277. <https://doi.org/10.1371/journal.pone.0032277>
- Felsenstein J (1985) Confidence limits on phylogenies: an approach using the bootstrap. Evolution 39:783–791. <https://doi.org/10.1111/j.1558-5646.1985.tb00420.x>
- Gillieatt BF, Coleman NV (2024) Unravelling the mechanisms of antibiotic and heavy metal resistance co-selection in environmental bacteria. FEMS Microbiol Rev. <https://doi.org/10.1093/femsre/fuae017>
- Gong F, Xin S, Liu X, He C, Yu X, Pan L, Zhang S, Gao H, Xu J (2024) Multiple biological characteristics and functions of intestinal biofilm extracellular polymers: friend or foe? Front Microbiol 15:1445630. <https://doi.org/10.3389/fmicb.2024.1445630>
- Grant JR, Stothard P (2008) The CGView server: a comparative genomics tool for circular genomes. Nucleic Acids Res 36:W181–W184. <https://doi.org/10.1093/nar/gkn179>
- Guo H, Xue S, Nasir M, Gu J, Lv J (2021) Impacts of cadmium addition on the alteration of microbial community and transport of antibiotic resistance genes in oxytetracycline contaminated soil. J Environ Sci 99:51–58. <https://doi.org/10.1016/j.jes.2020.04.015>
- Gupta K, Chatterjee C, Gupta B (2012) Isolation and characterization of heavy metal tolerant gram-positive bacteria with bioremediation properties from municipal waste rich soil of Kestopur canal (Kolkata), West Bengal, India. Biologia 67:827–836. <https://doi.org/10.2478/s11756-012-0099-5>
- Gurevich A, Saveliev V, Vyahhi N, Tesler G (2013) QUASt: quality assessment tool for genome assemblies. Bioinformatics 29:1072–1075. <https://doi.org/10.1093/bioinformatics/btt086>
- Han S, Lu Y, Xie J, Fei Y, Zheng G, Wang Z, Liu J, Lv L, Ling Z, Berglund B, Yao M (2021) Probiotic gastrointestinal transit and colonization after oral administration: a long journey. Front Cell Infect Microbiol 11:609722. <https://doi.org/10.3389/fcimb.2021.609722>
- Haque MM, Haque MA, Mosharaf MK, Islam MS, Islam MM, Hasan M, Molla AH, Haque MA (2022) Biofilm-mediated decolorization, degradation and detoxification of synthetic effluent by novel biofilm-producing bacteria isolated from textile dyeing effluent. Environ Pollut 314:120237
- Huo Q, Li R, Chen C, Wang C, Long T, Liu X (2023) Study on potential microbial community to the waste water treatment from bauxite desilication process. Environ Sci Pollut Res Int 30:15438–15453. <https://doi.org/10.1007/s11356-022-23150-1>
- Ikuta KS, Swetschinski LR, Aguilar GR, Sharara F, Mestrovic T, Gray AP, Weaver ND, Wool EE, Han C, Hayoon AG, Aali A (2022) Global mortality associated with 33 bacterial pathogens in 2019: a systematic analysis for the Global Burden of Disease Study 2019. Lancet 400:2221–2248. [https://doi.org/10.1016/S0140-6736\(22\)02185-7](https://doi.org/10.1016/S0140-6736(22)02185-7)
- Jones SE, Paynich ML, Kearns DB, Knight KL (2014) Protection from intestinal inflammation by bacterial exopolysaccharides. J Immunol 192:4813–4820. <https://doi.org/10.4049/jimmunol.1303369>
- Kalpana R, Maheshwaran M, Vimali E, Soosai MR, Shivamathi CS, Moorthy IG, Ashokkumar B, Varalakshmi P (2020) Decolorization of textile dye by halophilic *Exiguobacterium* sp. VK1: biomass and exopolysaccharide (EPS) enhancement for bioremediation of Malachite green. Chem Select 5:8787–8797. <https://doi.org/10.1002/slct.202001648>
- Kang Y, Wang J, Li Z (2024) Meta-analysis addressing the characterization of antibiotic resistome in global hospital wastewater. J Hazard Mater 466:133577. <https://doi.org/10.1016/j.jhazmat.2024.133577>
- Keilberg D, Ottemann KM (2016) How *Helicobacter pylori* senses, targets and interacts with the gastric epithelium. Environ Microbiol 18:791–806. <https://doi.org/10.1111/1462-2920.13222>
- Keynan Y, Weber G, Sprecher H (2007) Molecular identification of *Exiguobacterium acetylicum* as the aetiological agent of bacteraemia. J Med Microbiol 56:563–564. <https://doi.org/10.1099/jmm.0.46866-0>
- Khan MAS, Nahid ZI, Miah MI, Rahman SR (2022) Draft genome analysis of a multidrug-resistant *Pseudomonas aeruginosa* CMPL223 from hospital wastewater in Dhaka, Bangladesh. J Glob Antimicrob Resist 30:237–240. <https://doi.org/10.1016/j.jgar.2022.06.028>
- Klein EY, Van Boeckel TP, Martinez EM, Pant S, Gandra S, Levin SA, Goossens H, Laxminarayan R (2018) Global increase and geographic convergence in antibiotic consumption between 2000 and 2015. Proc Natl Acad Sci 115:E3463–E3470. <https://doi.org/10.1073/pnas.1717295115>
- Kneubehl AR, Iyer R (2023) Comparative genomics analysis of the aromatic and xenobiotic degradation capacities and heavy metal resistance in seven environmentally derived bacterial isolates. Water Air Soil Pollut 234:482. <https://doi.org/10.1007/s11270-023-06495-2>
- Kuebutornye FK, Lu Y, Abarike ED, Wang Z, Li Y, Sakyi ME (2020) *In vitro* assessment of the probiotic characteristics of three *Bacillus* species from the gut of *Nile tilapia*, *Oreochromis niloticus*. Probiotics Antimicrob Proteins 12:412–424. <https://doi.org/10.1007/s12602-019-09562-5>
- Kumari S, Kumar A, Lepcha A, Kumar R (2024) Cold-adapted *Exiguobacterium sibiricum* K1 as a potential bio-inoculant in cold regions: physiological and genomic elucidation of biocontrol and

- plant growth promotion. Gene 916:148439. <https://doi.org/10.1016/j.gene.2024.148439>
- Li W, Wang A (2021) Genomic islands mediate environmental adaptation and the spread of antibiotic resistance in multiresistant *Enterococci* - evidence from genomic sequences. BMC Microbiol 21:55. <https://doi.org/10.1186/s12866-021-02114-4>
- Li Y, Petrova OE, Su S, Lau GW, Panmanee W, Na R, Hassett DJ, Davies DG, Sauer K (2014) BdlA, DipA and induced dispersion contribute to acute virulence and chronic persistence of *Pseudomonas aeruginosa*. PLoS Pathog 10:e1004168. <https://doi.org/10.1371/journal.ppat.1004168>
- Liang Z, Zhang Y, He T, Yu Y, Liao W, Li G, An T (2020) The formation mechanism of antibiotic-resistance genes associated with bacterial communities during biological decomposition of household garbage. J Hazard Mater 398:122973. <https://doi.org/10.1016/j.jhazmat.2020.122973>
- Liu B, Zheng D, Jin Q, Chen L, Yang J (2019) VFDB 2019: a comparative pathogenomic platform with an interactive web interface. Nucleic Acids Res 47:D687–D692. <https://doi.org/10.1093/nar/gky1080>
- Lopez MC, Galan B, Carmona M, Navarro Llorens JM, Pereto J, Porcar M, Getino L, Olivera ER, Luengo JM, Castro L, Garcia JL (2021) Xerotolerance: a new property in *Exiguobacterium* genus. Microorganisms 9:2455
- Mahdi I, Fahsi N, Hijri M, Sobeh M (2022) Antibiotic resistance in plant growth promoting bacteria: a comprehensive review and future perspectives to mitigate potential gene invasion risks. Front Microbiol 13:999988. <https://doi.org/10.3389/fmicb.2022.999988>
- Mallick I, Bhattacharyya C, Mukherji S, Dey D, Sarkar SC, Mukhopadhyay UK, Ghosh A (2018) Effective rhizoinoculation and biofilm formation by arsenic immobilizing halophilic plant growth promoting bacteria (PGPB) isolated from mangrove rhizosphere: a step towards arsenic rhizoremediation. Sci Total Environ 610:1239–1250. <https://doi.org/10.1016/j.scitotenv.2017.07.234>
- Mandal S, Saha KK, Mandal NC (2021) Molecular insight into key eco-physiological process in bioremediating and plant-growth-promoting bacteria. Front Agron 3:664126. <https://doi.org/10.3389/fagro.2021.664126>
- Marfetan JA, Gallo AL, Farias ME, Vélez ML, Pescuma M, Ordoñez OF (2023) *Exiguobacterium* sp. as a bioinoculant for plant-growth promotion and Selenium biofortification strategies in horticultural plants. World J Microbiol Biotechnol 39:134. <https://doi.org/10.1007/s11274-023-03571-x>
- Markowicz A, Bondarczuk K, Cycón M, Sułowicz S (2021) Land application of sewage sludge: response of soil microbial communities and potential spread of antibiotic resistance. Environ Pollut 271:116317. <https://doi.org/10.1016/j.envpol.2020.116317>
- Maroof L, Khan I, Hassan H, Azam S, Khan W (2022) Microbial degradation of low density polyethylene by *Exiguobacterium* sp. strain LM-1K2 isolated from plastic dumped soil. World J Microbiol Biotechnol 38:197. <https://doi.org/10.1007/s11274-022-03389-z>
- Martino E, Franco B, Piccoli G, Stocchi V, Perotto S (2002) Influence of zinc ions on protein secretion in a heavy metal tolerant strain of the ericoid mycorrhizal fungus *Oidiodendron maius*. Mol Cell Biochem 231:179–185. <https://doi.org/10.1023/A:1014485420186>
- Matsumoto Y, Kaito C, Morishita D, Kurokawa K, Sekimizu K (2007) Regulation of exoprotein gene expression by the *Staphylococcus aureus* cvfB Gene. Infect Immun 75:1964–1972. <https://doi.org/10.1128/IAI.01552-06>
- Merlin TS, Umar M, Puthiyedathu ST (2022) Genomic insights into symbiosis and host adaptation of an ascidian-associated bacterium *Bacillus aryabhattai* MCCB 387. Symbiosis 87:129–142. <https://doi.org/10.1007/s13199-022-00860-1>
- Mokni-Tlili S, Hechmi S, Ouzari HI, Mechergui N, Ghorbel M, Jedidi N, Hassen A, Hamdi H (2023) Co-occurrence of antibiotic and metal resistance in long-term sewage sludge-amended soils: influence of application rates and pedo-climatic conditions. Environ Sci Pollut Res 30:26596–26612. <https://doi.org/10.1007/s11356-022-23802-2>
- Muhammad M, Wahab A, Waheed A, Mohamed HI, Hakeem KR, Li L, Li WJ (2024) Harnessing bacterial endophytes for environmental resilience and agricultural sustainability. J Environ Manag 368:122201. <https://doi.org/10.1016/j.jenvman.2024.122201>
- Murray CJ (2024) Findings from the global burden of disease study 2021. Lancet 403:2259–2262. [https://doi.org/10.1016/S0140-6736\(24\)00769-4](https://doi.org/10.1016/S0140-6736(24)00769-4)
- Murray CJ, Ikuta KS, Sharara F, Swetschinski L, Aguilar GR, Gray A, Han C, Bisignano C, Rao P, Wool E, Johnson SC (2022) Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. Lancet 399:629–655. [https://doi.org/10.1016/s0140-6736\(21\)02724-0](https://doi.org/10.1016/s0140-6736(21)02724-0)
- Nataraj BH, Ramesh C, Mallappa RH (2021) Characterization of antibiotic resistance and virulence traits present in clinical methicillin-resistant *Staphylococcus aureus* isolates. Curr Microbiol 78:2001–2014. <https://doi.org/10.1007/s00284-021-02477-x>
- Pal C, Bengtsson-Palme J, Rensing C, Kristiansson E, Larsson DGJ (2014) BacMet: antibacterial biocide and metal resistance genes database. Nucleic Acids Res 42:D737–D743. <https://doi.org/10.1093/nar/gkt1252>
- Pandey N (2020) *Exiguobacterium*. Beneficial microbes in agro-ecology. Elsevier, Amsterdam, pp 169–183. <https://doi.org/10.1016/B978-0-12-823414-3.00010-1>
- Paneri M, Sevta P (2023) Overview of antimicrobial resistance: an emerging silent pandemic. Glob J Med Pharm Biomed Update. <https://doi.org/10.1038/s41467-024-50457-z>
- Parrilli E, Tedesco P, Fondi M, Tutino ML, Giudice AL, de Pascale D, Fani R (2021) The art of adapting to extreme environments: the model system *Pseudoalteromonas*. Phys Life Rev 36:137–161. <https://doi.org/10.1016/j.plrev.2019.04.003>
- Patra M, Dubey SK (2024) Understanding the spread of antibiotic resistance in vegetables cultivated with sewage sludge: implications for food safety and human health. Environ Syst Res 13:1–24. <https://doi.org/10.1186/s40068-024-00347-6>
- Patra M, Pandey B, Dubey SK (2024a) Prevalence of diverse antimicrobial resistance genes and bacteria in sewage treatment plant-derived sludge environment. FEMS Microbes. <https://doi.org/10.1093/femsmc/xtae004>
- Patra M, Upadhyay SN, Dubey SK (2024b) Synchrotron induced X-ray fluorescence spectroscopy reveals heavy metal translocation in sludge amended soil–plant systems: assessment of ecological and health risks. Environ Geochem Health 46:399. <https://doi.org/10.1007/s10653-024-02174-1>
- Pavez VB, Pacheco N, Castro-Severyn J, Pardo-Esté C, Álvarez J, Zepeda P, Krüger G, Gallardo K, Melo F, Vernal R, Aranda C (2023) Characterization of biofilm formation by *Exiguobacterium* strains in response to arsenic exposure. Microbiol Spectrum 11:e02657–e02723. <https://doi.org/10.1128/spectrum.02657-23>
- Pinochet-Barros A, Helmann JD (2018) Redox sensing by Fe²⁺ in bacterial Fur family metalloregulators. Antioxid Redox Signal 29:1858–1871. <https://doi.org/10.1089/ars.2017.7359>
- Prabhu C, Satyaprasad AU, Deekshit VK (2024) Understanding bacterial resistance to heavy metals and nanoparticles: mechanisms, implications, and challenges. J Basic Microbiol. <https://doi.org/10.1002/jobm.202400596>
- Pu M, Rowe-Magnus DA (2018) A Tad pilus promotes the establishment and resistance of *Vibrio vulnificus* biofilms to mechanical clearance. Njp Biofilms Microbiomes 4:10. <https://doi.org/10.1038/s41522-018-0052-7>

- Read DS, Gweon HS, Bowes MJ, Anjum MF, Crook DW, Chau KK, Shaw LP, Hubbard A, AbuOun M, Tipper HJ, Hoosdally SJ (2024) Dissemination and persistence of antimicrobial resistance (AMR) along the wastewater-river continuum. *Water Res* 264:122204. <https://doi.org/10.1016/j.watres.2024.122204>
- Ren F, Li X, Tang H, Jiang Q, Yun X, Fang L, Huang P, Tang Y, Li Q, Huang J, Jiao XA (2018) Insights into the impact of flhF inactivation on *Campylobacter jejuni* colonization of chick and mice gut. *BMC Microbiol* 18:1–11. <https://doi.org/10.1186/s12866-018-1318-1>
- Ren A, Zhou Y, Xu Z, Jia T, Yang L (2024) Multiple-species biofilms as structuralized microbial communities for modulating microbiota homeostasis in human. *Curr Med* 3:12. <https://doi.org/10.1007/s44194-024-00039-4>
- Rihacek M, Kosaristanova L, Fialova T, Kuthanova M, Eichmeier A, Hakalova E, Cerny M, Berka M, Palkovicova J, Dolejska M, Svec P (2023) Zinc effects on bacteria: insights from *Escherichia coli* by multi-omics approach. *mSystems*. <https://doi.org/10.1128/msystems.00733-23>
- Rolff J, Bonhoeffer S, Kloft C, Leistner R, Regoes R, Hochberg ME (2024) Forecasting antimicrobial resistance evolution. *Trends Microbiol* 32:736–745. <https://doi.org/10.1016/j.tim.2023.12.009>
- Seemann T (2014) Prokka: rapid prokaryotic genome annotation. *Bioinformatics* 30:2068–2069. <https://doi.org/10.1093/bioinformatics/btu153>
- Seukep AJ, Mbuntcha HG, Kuete V, Chu Y, Fan E, Guo MQ (2022) What approaches to thwart bacterial efflux pumps-mediated resistance? *Antibiotics* 11:1287. <https://doi.org/10.3390/antibiotics11101287>
- Shafiq A, Khan HM, Shahid M, Fatima N, Khan MA (2024) Mutational insights into GyrA and GyrB genes in *Mycobacterium tuberculosis*: a genetic basis for fluoroquinolone resistance in multidrug-resistant tuberculosis. *J Pure Appl Microbiol* 18:2651–2658
- Shen Y, Ryser ET, Li H, Zhang W (2021) Bacterial community assembly and antibiotic resistance genes in the lettuce-soil system upon antibiotic exposure. *Sci Total Environ* 778:146255. <https://doi.org/10.1016/j.scitotenv.2021.146255>
- Shen L, Liu Y, Chen L, Lei T, Ren P, Ji M, Song W, Lin H, Su W, Wang S, Rooman M (2024) Genomic basis of environmental adaptation in the widespread poly-extremophilic *Exiguobacterium* group. *ISME J* 18:ward020. <https://doi.org/10.1093/ismejo/wrad020>
- Shippy DC, Eakley NM, Mikheil DM, Fadl AA (2014) Role of the flagellar basal-body protein, FlgC, in the binding of *Salmonella enterica* serovar Enteritidis to host cells. *Curr Microbiol* 68:621–628. <https://doi.org/10.1007/s00284-014-0521-z>
- Shree P, Singh CK, Sodhi KK, Surya JN, Singh DK (2023) Biofilms: understanding the structure and contribution towards bacterial resistance in antibiotics. *Med Microecol* 16:100084. <https://doi.org/10.1016/j.medmic.2023.100084>
- Sievers F, Higgins DG (2021) The clustal omega multiple alignment package. *Mult Seq Align Methods Protoc*. https://doi.org/10.1007/978-1-0716-1036-7_1
- Singh A, Rani PS, Bandsode V, Nyamero M, Kumar S, Ahmed N (2024a) Drivers of virulence and antimicrobial resistance in gram negative bacteria in different settings: a genomic perspective. *Infect Genet Evol*. <https://doi.org/10.1016/j.meegid.2024.105666>
- Singh CK, Sodhi KK, Shree P, Nitin V (2024b) Heavy metals as catalysts in the evolution of antimicrobial resistance and the mechanisms underpinning co-selection. *Curr Microbiol* 81:148. <https://doi.org/10.1007/s00284-024-03648-2>
- Su Z, Wang S, Yang S, Yin Y, Cao Y, Li G, Ma T (2021) Genetic and comparative genome analysis of *Exiguobacterium aurantiacum* SW-20, a petroleum-degrading bacteria with salt tolerance and heavy metal-tolerance isolated from produced water of Changqing oil field, China. *Microorganisms* 10:66. <https://doi.org/10.3390/microorganisms10010066>
- Sun Y, Zhang Y, Hao X, Zhang X, Ma Y, Niu Z (2023) A novel marine bacterium *Exiguobacterium marinum* a-1 isolated from in situ plastsphere for degradation of additive-free polypropylene. *Environ Pollut* 336:122390. <https://doi.org/10.1016/j.envpol.2023.122390>
- Sur S, Sathivelu M (2024) Functional profiling of the rhizospheric *Exiguobacterium* sp. for dimethoate degradation, PGPR activity, biofilm development, and ecotoxicological risk. *Sci Rep* 14:29361. <https://doi.org/10.1038/s41598-024-80559-z>
- Thomas GA, Gil TP, Müller CT, Rogers HJ, Berger CN (2024) From field to plate: how do bacterial enteric pathogens interact with ready-to-eat fruit and vegetables, causing disease outbreaks? *Food Microbiol* 117:104389. <https://doi.org/10.1016/j.fm.2023.104389>
- Thompson NT, Kitzenberg DA, Kao DJ (2023) Persister-mediated emergence of antimicrobial resistance in agriculture due to antibiotic growth promoters. *AIMS Microbiol* 9:738. <https://doi.org/10.3934/microbiol.2023038>
- Tian Q, Ye H, Zhou X, Wang J, Zhang L, Sun W, Duan C, Fan M, Zhou W, Bi C, Ye Q (2024) Evaluating the health risk of probiotic supplements from the perspective of antimicrobial resistance. *Microbiol Spectr*. <https://doi.org/10.1128/spectrum.00019-24>
- Ullah H, Hassan SH, Yang Q, Salama ES, Liu P, Li X (2025) Dynamic interaction of antibiotic resistance between plant microbiome and organic fertilizers: sources, dissemination, and health risks. *World J Microbiol Biotechnol* 41:4. <https://doi.org/10.1007/s11274-024-04214-5>
- Ullmann IF, Nygaard AB, Tunsjø HS, Charnock C (2020) Whole genome sequencing and antibiotic diffusion assays, provide new insight on drug resistance in the genus *Pedobacter*. *FEMS Microbiol Ecol*. <https://doi.org/10.1093/femsec/fiaa088>
- Uyaguari-Díaz MI, Croxen MA, Luo Z, Cronin KI, Chan M, Baticados WN, Nesbitt MJ, Li S, Miller K, Dooley D, Hsiao W (2017) Antibiotic resistance genes in agriculture and urban influenced watersheds in South-western British Columbia. *BioRxiv*. <https://doi.org/10.1101/104851>
- Vani S, Vadakkan K, Mani B (2023) A narrative review on bacterial biofilm: its formation, clinical aspects and inhibition strategies. *Futur J Pharm Sci* 9:50. <https://doi.org/10.1186/s43094-023-00499-9>
- Vishnivetskaya TA, Kathariou S, Tiedje JM (2009) The *Exiguobacterium* genus: biodiversity and biogeography. *Extremophiles* 13:541–555. <https://doi.org/10.1007/s00792-009-0243-5>
- Wardal E, Żabicka D, Skalski T, Kubiak-Pulkowska J, Hryniewicz W, Sadowy E (2023) Characterization of a Tigecycline-, Linezolid- and vancomycin-resistant clinical *Enterococcus faecium* isolate, carrying vanA and vanB genes. *Infect Dis Ther* 12:2545–2565. <https://doi.org/10.1007/s40121-023-00881-3>
- Wernli D, Harbarth S, Levrat N, Pittet D (2022) A ‘whole of United Nations approach’ to tackle antimicrobial resistance? A mapping of the mandate and activities of international organisations. *BMJ Glob Health* 7:e008181. <https://doi.org/10.1136/bmjgh-2021-008181>
- Wu Q, Tian A, Xu J, Fang Q, Huang H, Hu Y (2024) *Edwardsiella piscicida* requires SecY homeostasis facilitated by FtsH and YccA for stress resistance and virulence. *Aquaculture* 582:740528. <https://doi.org/10.1016/j.aquaculture.2023.740528>
- Xiao M, Long ZE, Fu X, Zou L (2024) Versatile applications and mechanisms of genus *Exiguobacterium* in bioremediating heavy metals and organic pollutants: a review. *Int Biodeterior Biodegrad* 194:105884. <https://doi.org/10.1016/j.ibiod.2024.105884>
- Xu X, Lin D, Yan G, Ye X, Wu S, Guo Y, Zhu D, Hu F, Zhang Y, Wang F, Jacoby GA (2010) vanM, a new glycopeptide resistance gene cluster found in *Enterococcus faecium*. *Antimicrob Agents Chemother* 54:4643–4647. <https://doi.org/10.1128/aac.01710-09>

- Xu L, Dong Z, Fang L, Luo Y, Wei Z, Guo H, Zhang G, Gu YQ, Coleman-Derr D, Xia Q, Wang Y (2019) OrthoVenn2: a web server for whole-genome comparison and annotation of orthologous clusters across multiple species. *Nucleic Acids Res* 47:W52–W58. <https://doi.org/10.1093/nar/gkz333>
- Yan Q, Xu Y, Zhong Z, Xu Y, Lin X, Cao Z, Feng G (2023) Insights into antibiotic resistance-related changes in microbial communities, resistome and mobilome in paddy irrigated with reclaimed wastewater. *Sci Total Environ* 900:165672. <https://doi.org/10.1016/j.scitotenv.2023.165672>
- Zhang D, Zhu Z, Li Y, Li X, Guan Z, Zheng J (2021) Comparative genomics of *Exiguobacterium* reveals what makes a cosmopolitan bacterium. *mSystems* 6:10–1128. <https://doi.org/10.1128/mSystems.00383-21>
- Zhang L, Adyari B, Hou L, Yang X, Gad M, Wang Y, Ma C, Sun Q, Tang Q, Zhang Y, Yu CP (2024) Mass-immigration shapes the antibiotic resistome of wastewater treatment plants. *Sci Total Environ* 908:168193. <https://doi.org/10.1016/j.scitotenv.2023.168193>
- Zheng S, Bawazir M, Dhall A, Kim HE, He L, Heo J, Hwang G (2021) Implication of surface properties, bacterial motility, and hydrodynamic conditions on bacterial surface sensing and their initial adhesion. *Front Bioeng Biotechnol* 9:643722. <https://doi.org/10.3389/fbioe.2021.643722>
- Zhou X, Zhang B, Li L (2024) From waste to resource: assessing the feasibility of municipal sludge as a fertilizer from a soil and microbial perspective. *Chem Eng J Adv* 19:100630. <https://doi.org/10.1016/j.cej.2024.100630>
- Zulkeflee SNM, Mustafa N, Hara H (2019) Identification and assessment of fluoroquinolone multidrug efflux pump in multidrug resistant *Staphylococcus aureus* isolated from hospital wastewater in Kuala Lumpur. *Int J Integr Eng* 11:7–13. <https://doi.org/10.30880/ijie.2019.11.07.002>

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